

Q1 The negation of "Some students like hockey" is:

- (A) Some students dislike hockey
- (B) Every student dislike hockey
- (C) Every student like hockey
- (D) All students like hockey

Q2 A relation 'R' is defined on ordered pairs of integers as $(x, y) R(u, v)$ if $x < u$ and $y > v$. Then R is

- (A) Neither a partial order nor an equivalence relation
- (B) A partial order but not a total order
- (C) A total order
- (D) An equivalence relation

Q3 Suppose you are married and you and your partner attend a party with three other married couples. Several handshakes took place. No one shook hands with himself (or herself) or with their partner, and no one shook hands with the same person more than once. After all handshaking was completed, suppose you asked each person, including your partner, how many hands they had shaken. Each person gave a different answer. How many hands did your spouse shake?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q4 Consider the following conditional code, which returns a Boolean values

```
if((x>25) && (y>100))
    return 'false';
else if((x<=25) && (y<=100))
    return 'true';
else if ( x>25) && (y<=100))
    return 'false';
else
    return 'true';
```

Simplify it by filling in the following blank with a single Boolean expression without changing the behaviour of the conditional code.

```
if ( -----)
    return 'true';
else
    return 'false';
```

- (A) $x > 25$
- (B) $x \leq 25$

(C) $y > 100$

(D) $y \leq 100$

Q5 Which one of the following types of memory is the fastest?

- (A) Cache Memory
- (B) Register Memory
- (C) Main Memory
- (D) Secondary Memory

Q6 The memory size for n address lines and m data lines is given by

- (A) $2^m \times n$
- (B) $m \times n^2$
- (C) $2^n \times m$
- (D) $n \times m^2$

Q7 Simplify the following using K-Map

$$F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$$

- (A) $BD + B'D'$
- (B) $AC + A'C'$
- (C) $BC + B'C'$
- (D) $AD + A'D'$

Q8 Consider an unpipelined machine with 10 nsec clock cycles which uses four cycles for ALU operations and branches where as five cycles for memory operation. Assume that the relative frequencies of these operations are: 40 %, 20 % and 40 %, respectively. Due to clock skew and setup pipeline let us consider that the machine adds one nsec overhead to the clock. How much speedup is observed in the instruction execution rate when a pipelined machine is considered?

- (A) 2 times
- (B) 4 times
- (C) 6 times
- (D) 8 times

Q9 Machine Level Language is a/an

- (A) Assembly Language
- (B) Low-Level language
- (C) High-level Language
- (D) Translating Language

Q10 Which of the following is wrong about the data types?

- (A) The number is always positive when qualifier 'unsigned' is used.
- (B) The number can be positive or negative when the qualifier 'signed' is used
- (C) The range of values for signed data types is more than that of unsigned data types
- (D) The left most bit in unsigned data type is used to represent the value



Q11 If a constructor 'Date' is declared explicitly and has to be defined outside the class, which of the following is correct?

- (A) `Date::Date(int dd) {/ *... * /}`
- (B) explicit `Date:: Date(int dd) {/ *... * /}`
- (C) Such a constructor cannot be defined
- (D) Constructor always has to be defined inside the class

Q12 Which one of the following allows the session to continue?

- (A) When a user quits a browser
- (B) When the user logs out and is invalidated by the servlet
- (C) When the session is timed out due to inactivity
- (D) When the user refreshes the browser and there is a persistent cookie

Q13 Overfitting is expected when we observe that?

- (A) With training iterations errors on the training set as well as the test set decrease
- (B) With training iterations error on the training set decreases but the test set increases
- (C) With training iterations errors on the training set as well as the test set increase
- (D) With training iterations training set as well as test error remain constant

Q14 Identify the incorrect statement(s).

- (A) A candidate key is minimal set of one or more attributes that, taken collectively, allows us to uniquely identify any entity in the entity set.
- (B) A candidate key for which no proper subset is also a candidate key is called a super key.
- (C) A super key is a set of one or more attributes that, taken collectively, allows us to uniquely identify any entity in the entity set.
- (D) A super key for which no proper subset is also a super key is called a candidate key.

Q15 Consider the two relations below. The primary keys are underlined. Identify all possible foreign key(s) from the options based only on the two relations.

EMP (eid, ename, did)

DEPT (did, dname)

- (A) eid
- (B) did
- (C) eid, did

(D) eid, did, ename

Q16 Which of the following statements is correct, according to the following instance of the relational schema $R(X, Y, Z)$?

X	Y	Z
1	1	a
1	1	a
2	1	b
3	2	b

- (A) $X \rightarrow Y, Z \rightarrow X$
- (B) $Y \rightarrow Z, Z \rightarrow X$
- (C) $X \rightarrow Y, X \rightarrow Z$
- (D) $Y \rightarrow X, X \rightarrow Z$

Q17 In Linux Operating System, when _____ is invoked, it is passed a set of flags that determine how much sharing is to take place between the parent and child tasks.

- (A) `fork()`
- (B) `clone()`
- (C) `pthread()`
- (D) `thread()`

Q18 The main function of the microkernel is to provide a communication facility between the _____ program and the various _____ that are also running in user space.

- (A) Virtual, Processes
- (B) System, Processes
- (C) Client, Services
- (D) Virtual, Services

Q19 Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is first come first served (FCFS). If FCFS is replaced by shortest seek time first (SSTF) and the vendor claims 50 % better benchmark results. What is the expected improvement in the I/O performance of user programs?

- (A) 50%
- (B) 100%
- (C) 25%
- (D) 0%

Q20 For the following set of processes scheduled using FCFS policy, determine the average waiting time. Assume that the processes arrived in the order P1, P2, P3, P4.

Process	Burst Time
P1	8



P2	15
P3	10
P4	7

- (A) 8 (B) 16
(C) 32 (D) 48

Q21 Which of the following is not a Non-functional requirement?

- (A) Portability.
(B) Security.
(C) Scalability
(D) User interaction

Q22 COCOMO stands for-

- (A) Consumed cost model
(B) Constructive cost model
(C) Common control model
(D) Composition cost model

Q23 Modifying the software to match changes in the ever-changing environment is called as

- (A) Adaptive maintenance
(B) Corrective maintenance
(C) Perfective maintenance
(D) Preventive maintenance

Q24 For a function of two variables, boundary value analysis yields,

- (A) $4n+3$ test cases
(B) $4n+1$ test cases
(C) $n+4$ test cases
(D) $2n+4$ test cases

Q25 How many different binary trees are possible with 'n' nodes?

- (A) $n-1$ (B) 2^n-1
(C) 2^n (D) $\frac{1}{n+1} \binom{2n}{n}$

Q26 Let 'n' denote a positive integer. Suppose a function F is defined as

$$f(n) = \begin{cases} 0, & n = 1 \\ f\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right), & n > 1 \end{cases}$$

What is $f(25)$? and what does this function find?

- (A) 4, $\lfloor \log_2 n \rfloor$
(B) 14, $\lfloor \log_2 n \rfloor$
(C) 4, $\lfloor \frac{n}{2} \rfloor$
(D) 14, $\lfloor \frac{n}{2} \rfloor$

Q27 Given the FFT we can have _____ time procedure for multiplying two polynomials $A(x)$

and $B(x)$ of degree bound n where input and output representations are in coefficient form, assuming n is a power of 2.

- (A) $O(n^2)$ (B) $O(n \log_2 n)$
(C) $O(2^n)$ (D) $O(\log_2 n)$

Q28 Consider the Graph below:



How many spanning trees can be found?

- (A) 10 (B) 5
(C) 9 (D) 8

Q29 The transition function ' δ ' in multi-tape Turing machine is defined as:

- (A) $\delta : 2Q \times \Gamma^k \rightarrow 2^Q \times \Gamma^k \times \{L, R, S\}^k$
(B) $\delta : Q \times Q \times \Gamma^k \rightarrow Q \times Q \times \Gamma^k \times \{L, R, S\}^k$
(C) $\delta : Q \times \Gamma^k \rightarrow Q \times \Gamma^k \times \{L, R, S\}^k$
(D) $\delta : Q \times \Gamma^k \times 2^Q \rightarrow Q \times \Gamma^k \times 2^Q \times \{L, R, S\}^k$

Q30 In Pumping Lemma for regular languages, to say a language is satisfying pumping lemma, what is the minimum length of 'y' if you consider the string as 'x y z'?

- (A) n (B) 2
(C) 1 (D) 0

Q31 The item generated by the production $A \rightarrow \epsilon$ is :

- (A) $A \rightarrow \cdot$ (B) $A \rightarrow \epsilon \cdot$
(C) $A \rightarrow \cdot \epsilon$ (D) $A \rightarrow \epsilon$

Q32 In the ϵ -NFA, $M = (\{q_0, q_1, q_2, q_3\}, \{a\}, \delta, q_0, \{q_3\})$ where ' δ ' is given in the transition table below, what is the minimum length of string to reach to the final state?

	ϵ	a
q_0	$\{q_1\}$	$\{\}$
q_1	$\{q_2\}$	$\{\}$
q_2	$\{\}$	$\{q_2, q_3\}$
q_3	$\{\}$	$\{\}$

- (A) 0 (B) 1
(C) 2 (D) 3

Q33



Given Ethernet address

010110100001000101010101000110001010101000001111

in binary, what is the address in hexadecimal notation?

- (A) 5A:88:AA:18:55:F0
- (B) 5A: 81: BA: 81: AA: 08
- (C) 5 A: 18: 5A:18: 55: 0F
- (D) 5A:11:55:18:AA:0F

- Q34** A Computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. It is initially filled to capacity with 16 megabits. What is the maximum duration for which the computer can transmit at the full 10 Mbps ?
- (A) 1.6 seconds
 - (B) 2 seconds
 - (C) 5 seconds
 - (D) 8 seconds

- Q35** In the standard Ethernet with transmission rate of 10 Mbps, assume that the length of the medium is 2500 m and size of a frame is 512 bytes. The propagation speed of a signal in a cable is normally 2×10^8 m/s. Calculate Transmission delay and propagation delay.
- (A) 25. $25\mu s$ and 51. $2\mu s$
 - (B) 51. $2\mu s$ and 12. $5\mu s$
 - (C) 10. $24\mu s$ and 50. $12\mu s$
 - (D) 12. $5\mu s$ and 51. $2\mu s$

- Q36** In CRC coding if the data word is 111111, divisor is 1010 and the remainder is 110. Which of the following code is true?
- (A) 011111101
 - (B) 001010110
 - (C) 111111110
 - (D) 110111111

- Q37** The A* algorithm is optimal when,
- (A) It always finds the solution with the lowest total cost if the heuristic 'h' is admissible.
 - (B) Always finds the solution with the highest total cost if the heuristic 'h' is admissible.
 - (C) Finds the solution with the lowest total cost if the heuristic 'h' is not admissible.
 - (D) It always finds the solution with the highest total cost if the heuristic 'h' is not admissible.

- Q38** Which Artificial intelligence technique enables computers to understand the associations and relationships between objects & Events?
- (A) Heuristic Processing
 - (B) Cognitive Science

- (C) Relative symbolism
- (D) Pattern Matching

- Q39** Where does the values of alpha-beta search get updated?
- (A) Along the path of search.
 - (B) Initial state itself.
 - (C) At the end.
 - (D) None of these.

- Q40** A 4-input neuron has weights 1,2,3,4. The transfer function is linear with the constant of proportionality being equal to 3. The inputs are 5,7,10,30, respectively. Then the output will be
- (A) 120
 - (B) 213
 - (C) 410
 - (D) 507

- Q41** Consider the following statements:
- P: There exists no simple, undirected and connected graph with 80 vertices and 77 edges.
- Q: All vertices of Euler graph are of even degree.
- R: Every simple, undirected, connected and acyclic graph with 50 vertices has at least two vertices of degree one.
- S: There exists a bipartite graph with more than ten vertices which are 2 -2-colorable.
- What is the number of correct statements among the above statements?
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4

- Q42** Consider the following statements:
- A. There exists a Boolean algebra with ' 5 ' elements.
- B. Every element of Boolean algebra has a unique complement.
- C. If a Lattice ' L ' is a Boolean algebra then ' L ' is not relatively complemented.
- D. The direct product of two Boolean Algebras is also a Boolean algebra.
- Choose the correct answer about the four statements given above.
- (A) Only A and D are correct
 - (B) Only B and D are correct
 - (C) All statements are NOT correct
 - (D) All statements are correct

- Q43** Which of the statement is not true in regards to virtual memory?



- (A) The main objective for using virtual memory is to increase the effective capacity of the memory system
- (B) Size of the program can be as large as the size of the secondary memory
- (C) Program and data are stored in the secondary memory
- (D) None of these

Q44 Which of the following is/are true for Dynamic RAM (DRAM)?

1. It is slower than static RAM(SRAM)
2. Packing Density is higher than Static RAM (SRAM)
3. It is faster than static RAM(SRAM)
4. It requires data refreshing

Choose the correct answer from the options given below:

- (A) 1, 2, and 4 Only
- (B) 1, 2, and 3 Only
- (C) 3 and 4 Only
- (D) 2 and 4 Only

Q45 Which of the following statement is/are false?

1. The processor has direct access to both primary and secondary memory
2. Primary memory stores the active instructions and data for the program being executed on the process
3. Secondary memory is used as a backup memory
4. Memory system is implemented on a single-level memory

Choose the correct answer from the options given below:

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1, 2 and 3 only
- (D) 1 and 4 only

Q46 Given a vector with cartesian components (x,y), if scaling is done with matrix $\begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$, which of the following are true.

1. Decreases the vertical by three halves
2. Increases the vertical by three halves
3. Doubles the horizontal
4. Halves the horizontal

Choose the correct answer from the options given below:

- (A) 1 and 3 only (B) 1 and 4 only
- (C) 2 and 3 only (D) 2 and 4 only

Q47 Which of the following lists defined in HTML share a similar element

1. Unordered lists
2. Ordered lists
3. Definition Lists
4. Correlated lists
5. Enumerated lists

Choose the most appropriate answer from the options given below:

- (A) 1, 2 only (B) 1, 3, 5 only
- (C) 1, 2, 3 only (D) 1, 3, 4 only

Q48 What are the drawbacks of using file systems to store data?

1. Data inconsistency
2. Difficulty in accessing data
3. Data isolation
4. Atomicity of updates

Choose the correct answer from the options given below:

- (A) 1,2 only (B) 2,3,4 only
- (C) 1,2,3 only (D) 1,2,3,4

Q49 Which of the following database model, we have a parent-child relationship?

1. hierarchical databases
2. network databases
3. relational databases

Choose the correct answer from the options below:

- (A) 1,2 (B) 1,3
- (C) 2,3 (D) 1,2,3

Q50 Identify the correct operation that produces the below-given output based on two relations R1 and R2.

R1

Name	Class
a1	9
b1	5
c1	6
d1	11

R2

Name	Class
------	-------



b1	5
f1	8

Output:

Name	Class
b1	5

Choose the correct answer from the options below:

- (A) $R1 \cap R2$ (B) $R1 - R2$
(C) $R1 - (R1 - R2)$ (D) $R1 \cup R2$

Q51 Which of the following statements are true?

1. Shortest remaining time first scheduling may cause starvation
2. Preemptive scheduling may cause starvation
3. Round robin is better than FCFS in terms of response time

Choose the correct answer from the options given below:

- (A) 1 only (B) 2,3 only
(C) 1,2 only (D) 1,2,3

Q52 In the design protocol of the critical section problem, each process must ask permission to enter the critical section in _____ code; it then executes in the critical section; once it finishes executing in the critical section it enters the _____ code. The process then enters the _____ code.

- (A) entry section, remainder section, exit section
(B) entry section, exit section, remainder section
(C) remainder section, entry section, exit section
(D) remainder section, exit section, entry section

Q53 How do you create agile processes to manage unpredictability?

1. Requirements gathering must be conducted very carefully.
2. Risk analysis must be conducted before planning takes place.
3. Software increments must be delivered in short time periods.
4. Software processes must not adapt to changes incrementally.

Choose the correct answer from the options given below:

- (A) 1 only (B) 2 only
(C) 3 only (D) 4 only

Q54

Software project manager is responsible for the following tasks:

- A. Project Planning
B. Project status tracking
C. Resource management
D. Risk management
E. Project delivery within time & budget.

Choose the correct answer from the options given below:

- (A) All the statements are correct.
(B) Only B & C are correct.
(C) Only A & D are correct
(D) All the statements are not correct.

Q55 Mr. X designed open source software which must comply with some criteria. Choose the false statement in respect of the above.

1. No restriction on redistribution of the software as part or whole.
2. The integrity of the author's source code must be maintained.
3. The software can be sold after distribution.

Choose the correct answer from the options given below:

- (A) 1 only
(B) 2 only
(C) 3 only
(D) All statements are true.

Q56 Circuit satisfiability problem: Given a Boolean combinatorial circuit composed of AND, OR, and NOT gates, is it satisfiable? A one-output Boolean combinatorial circuit is satisfiable means for the given inputs the output will be 1.

1. The circuit satisfiability problem belongs to class NP
2. The circuit satisfiability problem is at least as hard as any language in NP
3. The circuit satisfiability is NP-Complete
4. The size of the circuit is $\theta(K^2+1)$
5. If $P \neq NP$, this situation would contradict the NP-Completeness of the problem.

Choose the correct answer from the options given below:

- (A) 1, 3, 4 Only (B) 2, 4, 5 Only
(C) 1, 2, 3 Only (D) 2, 3, 4 Only

Q57 Given an expression $(A+B * D) / (E-F)+G$



1. The prefix notation is $+/*$ DB-EFG
2. The prefix expression is the reverse of the postfix expression
3. The order of operands in infix expression and postfix expression are the same.
4. The order of operands in infix expression, prefix expression, and postfix expression are the same.

Choose the most appropriate answer from the options given below:

- (A) 1, 3, and 4 Only
(B) 3 and 4 Only
(C) 1 and 2 Only
(D) 1 and 4 Only

Q58 Which of the following is (are) correct about the regular expression?

$a^* b^* c^* d^*$

1. The language for the given expression is:

L

$= \{a^n b^m c^m d^m \mid n$

$\geq 1, m \geq 1\} \cup \{a^n b^m c^m d^n \mid n \geq 1, m \geq 1\}$

2. The Context Free Language for the given expression is:

$S \rightarrow AB|C$

$A \rightarrow aAb|ab$

$B \rightarrow cBd|cd$

$C \rightarrow aCd|aDd$

$D \rightarrow bDc|bc$

3. The language generated by this expression is equal number of 'a's, followed by equal number of 'b's, followed by equal number of 'c's and followed by equal number of 'd's.

Choose the correct answer from the options given below:

- (A) Only 1 is correct
(B) Only 2 is correct
(C) Both 1 and 2 are correct
(D) All the three 1, 2 and 3 are correct

Q59 Which of the following statements are correct?

1. The first three-address instruction in the intermediate code is a leader.
2. The instruction which is exactly at the middle of the given three address code is a leader.
3. Any instruction that is the target of a conditional or unconditional jump is a leader.
4. Any instruction that immediately follows a conditional or unconditional jump is a leader.

Choose the correct answer from the options given below:

- (A) 1, 2 and 3
(B) 2, 3 and 4
(C) 1,3,4
(D) 1,2,3,4

Q60 Which of the following are correct on regular expressions?

a. $\varphi + L = L + \varphi = L$

b. $\varepsilon L = L\varepsilon = L$

c. $\varphi L = L\varphi = \varphi$

d. $\varphi L = L\varphi = L$

Choose the correct answer from the options given below:

- (A) a, b and d only
(B) a, b and c only
(C) b and d only
(D) a and d only

Q61 Consider two hosts P and Q are connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes and between R and Q is 820 bytes. A TCP segment of size of 1400 bytes is transferred from P to Q through R with IP identification value of 0x1234. Assume that IP header size is 20 bytes. Further, the packet is allowed to be fragmented that is Don't Fragment (DF) flag in the IP Header is not set by P. Which of the following statements is/are true

- I. Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes
- II. If the second fragment is lost then R resends the fragment with IP Identification value of 0x1234
- III. If the second fragment is lost then P is required to resend the entire TCP segment
- IV. TCP destination port can be determined by analyzing the second fragment only

Choose the correct answer from the options given below:

- (A) I, II, and III only
(B) I and III only
(C) III and IV only
(D) II and IV only

Q62



The Solution to Silly Window Syndrome problem is/are:

- I. Nagle's Algorithm
- II. Clark's Algorithm
- III. Jacobson's Algorithm
- IV. Piggy backing Algorithm

Choose the correct answer from the options given below:

- (A) I and II only
- (B) I and III only
- (C) III and IV only
- (D) II and IV only

Q63 Which AI System mimics the evolutionary process to generate increasingly better solutions to a process to a problem?

- I. Self organizing neural network.
- II. Back propagation neural network.
- III. Genetic algorithm.
- IV. Forward propagation neural network.

Choose the correct answer from the options given below:

- (A) I only
- (B) II only
- (C) III only
- (D) IV only

Q64 Consider the following Learning algorithms.

- I. Logistic regression.
- II. Back propagation.
- III. Linear regression.
- IV. Forward propagation.

Which of the following options represents classification algorithm?

Choose the correct answer from the options given below:

- (A) I and III only
- (B) II and IV only
- (C) I and II only
- (D) III and IV only

Q65 Which Boolean operation on two variables can be represented by a single perception layer?

- I. $X1 \text{ AND } X2$
- II. $X1 \text{ OR } X2$
- III. $X1 \text{ NOR } X2$
- IV. $X1 \text{ XOR } X2$

Choose the most appropriate answer from the options given below:

- (A) I and II only
- (B) II and III only

- (C) I, II, and III only
- (D) I, II, III, and IV only

Q66 Match the List-I and List-II

List-I	List-II
a. Planner Graph	I. Probabilistic Model
b. Bipartite Graph	II. Deterministic Model
c. PERT	III. 4-Colorable
d. CPM	IV. 2-Colorable

Choose the correct answer from the options given below:

- (A) a-IV; b-III; c-I; d-II
- (B) a-III; b-IV; c-II; d-I
- (C) a-II; b-IV; c-I; d-III
- (D) a-III; b-IV; c-I; d-II

Q67 Match List-I and List-II

List-I	List-II
a. Direct Addressing	I. MVI A, FFH
b. indirect addressing	II. MOV A, B
c. Register Addressing	III. STA, FFFFA
d. Immediate addressing	IV. MOV A, M

Choose the correct answer from the options given below:

- (A) a-I, b-II, c-III, d-IV
- (B) a-II, b-III, c-IV, d-I
- (C) a-III, b-IV, c-II, d-I
- (D) a-IV, b-III, c-I, d-II

Q68 Match the List-I and List-II

List-I	List-II
a. If the implementation of generated or derived classes differs only through a parameter we have	I. to use class hierarchy
b. If the actual types of objects used cannot be known at compile time, then we have	II. improved run-time efficiency
c. If inline operations are essential and templates are used then we have	III. to use templates
d. To gain access to differing instances for	IV. to use explicit casting



derived classes through
base we have

Choose the correct answer from the options
given below:

- (A) a-I, b-II, c-IV, d-III
(B) a-III, b-I, c-II, d-IV
(C) a-I, b-III, c-II, d-IV
(D) a-II, b-I, c-III, d-IV

Q69 Match List I and List II

List-I	List-II
a. Apache Kafka	I. Platform for constructing data flows for extract, transform, and load (ETL) processing and analysis of large datasets
b. Pig	II. General-purpose computing model and runtime system for distributed data analytics.
c. Apache Mahour	III. Open-source platform that was created by LinkedIn
d. Mapreduce	IV. Open-source platform used for creating scalable machine learning algorithms

Choose the correct answer from the options
given below:

- (A) a-II, b-IV, c-I, d-III
(B) a-II, b-I, c-IV, d-III
(C) a-III, b-I, c-IV, d-II
(D) a-I, b-II, c-IV, d-III

Q70 Match List I with List II

List-I	List-II
a. IPC	I. Resource Allocation
b. Demand Paging	II. Computational speedup
c. Banker's Algorithm	III. Task Control Block
d. PCB	IV. Virtual Memory

Choose the correct answer from the options
given below:

- (A) a-II, b-I, c-IV, d-III
(B) a-II, b-IV, c-I, d-III

- (C) a-I, b-II, c-III, d-IV
(D) a-II, b-III, c-I, d-IV

Q71 Match List I with List II

List-I	List-II
a. Product Complexity	I. System Requirements Design
b. Structured System analysis	II. Software design
c. Coupling & Cohesion	III. Validation techniques
d. Symbolic Execution	IV. Software cost estimation

Choose the correct answer from the options
given below:

- (A) a-II, b-III, c-IV, d-I
(B) a-III, b-I, c-IV, d-II
(C) a-IV, b-I, c-II, d-III
(D) a-III, b-IV, c-I, d-II

Q72 Match List I with List II

List-I	List-II
a. Dijkstra's Algorithm	I Calculates path matrix
b Prim's Algorithm	II Stores minimum cost edge
c Warshall's Algorithm	III Stores the total cost from a source node to the current node
d Kruskal's algorithm	IV Finds Minimum Spanning Tree

Choose the correct answer from the options
given below:

- (A) a-I, b-II, c-III, d-IV
(B) a-III, b-II, c-I, d-IV
(C) a-II, b-I, c-IV, d-III
(D) a-III, b-IV, c-II, d-I

Q73 Match the following based on the language
accepted by using brute force method of parsing.

List-I	List-II
a. $S \rightarrow aSa aa$	I. $((2^n * 3) - 4); n \geq 1$
b. $S \rightarrow aaSaa aa$	II. $2^n; n \geq 1$
c. $S \rightarrow aaaSaaa aa$	III. $((4 * 2^n) - 6); n \geq 1$
d. $S \rightarrow aaaaSaaaa aa$	IV. $2^n - 2; n \geq 2$

Choose the correct answer from the options
given below:

- (A) a-IV, b-III, c-I, d-II
(B) a-II, b-IV, c-I, d-III



- (C) a-I, b- IV, c-III, d-II
 (D) a-II, b-III, c-I, d-IV

Q74 Match List I with List II

List-I	List-II
a.CDMA	I. It provides mobile internet connection with faster data transfer rates
b.GSM	II. It allows users to connect to a network or to other devices over wireless channel
c.UMTS	III. Accessing mechanism for multiple transmitters over a single channel
d. WiFi	IV. It is cellular technology employs hybrid of FDMA and TDMA

Choose the correct answer from the options given below:

- (A) a-III, b-IV, c-II, d-I
 (B) a-III, b-IV, c-I, d-II
 (C) a-II, b-III, c-IV, d-I
 (D) a-II, b-I, c-IV, d-III

Q75 Match List I with List II

List-I	List-II
a.Text planning	I. Natural language understanding
b.Sentence planning	II. Natural language generation
c.Sentence generation	
d.Map the input to useful representations	

Choose the correct answer from the options given below:

- (A) a-I, b-II, c-I, d-II
 (B) a-II, b-II, c-I, d-II
 (C) a-I, b-II, c-II, d-I
 (D) a-II, b-II, c-II, d-I

Q76 Consider the statements:

I. There does not exist a polynomial time algorithm to solve integer linear programming problems.

II. Main focus of PERT is 'minimizing time'.

Choose the correct option about statements I and II.

- (A) I is True; II is True
 (B) I is True; II is False
 (C) I is False; II is True
 (D) I is False; II is False

Q77 The correct sequence in the fetch-execute cycle is _____ .

- I. Decode
 II. Fetch
 III. Execute

Choose the correct answer from the following

- (A) I-II-III (B) II-III-I
 (C) III-II-I (D) II-I-III

Q78 In what sequence the initialization, testing and execution of body is done while using do-while loop?

- I. Commenting
 II. Execution of the body
 III. Initialisation
 IV. Testing the condition

Choose the correct answer from the following

- (A) IV, II, III (B) IV, III, II
 (C) III, I, II (D) III, II, IV

Q79 Select the correct order of DBSCAN algorithm.

I. Find recursively all its density-connected points and assign them to the same cluster as the core point.

II. Find all the neighbor points with EPS and identify the core points with more than MinPts neighbors.

III. Iterate through the remaining unvisited points in the dataset.

IV. For each core point if it is not already assigned to a cluster, create a new cluster.

Choose the correct answer from the following :

- (A) II, IV, III, I (B) IV, II, III, I
 (C) II, IV, I, III (D) IV, II, I, III

Q80 Select the correct order of events after power is initialized on a system.

- I. Bootstrap loader is loaded from the disk
 II. Kernel is loaded onto the memory
 III. Firmware ROM loads boot block

Choose the correct answer from the options given below:

- (A) II, III, and I (B) III, I, and II
 (C) I, II, and III (D) I, III, and II



- Q81** Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R.

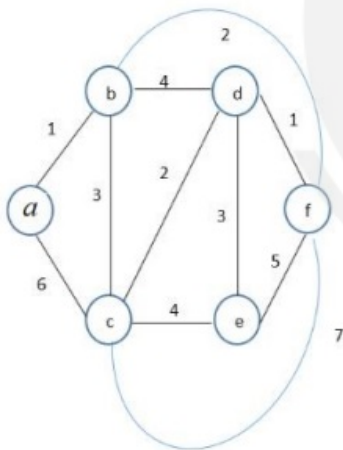
Assertion A: Cohesion is a qualitative indication of the degree to which a module can be written more compactly and is able to complete its function in a timely manner.

Reason R: Cohesion is a qualitative indication of the degree to which a module is connected to other modules and the outside world.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both A and R are correct and R is the correct explanation of A.
- (B) Both A and R are correct but R is NOT the correct explanation of A.
- (C) A is correct but R is not correct.
- (D) A is not correct but R is correct.

- Q82** Given the graph below one of the following edges cannot be added in that order to find a minimum spanning tree algorithm.



- I. a-b
- II. d-f
- III. b-f
- IV. d-c
- V. d-e

Choose the correct answer from the options given below:

- (A) I, II, III, IV, V
- (B) I, II, IV, III, V
- (C) II, I, III, V, IV
- (D) II, I, IV, III, V

Q83

What is the safest order while simplifying Context Free Grammar?

- (A) Elimination of $\in$ -productions, Unit productions, and then Useless symbols & productions.
- (B) Elimination of useless symbols productions, $\in$ -productions, and then Unit productions.
- (C) Elimination of Unit productions, $\in$ -productions and then Useless symbols and productions.
- (D) Elimination of $\in$ -productions, Useless symbols and productions, and then Unit productions.

- Q84** Rearrange the following sequence in the context of OSI Layers:

- I. Transforming the raw bits in the form of frame for transmission
- II. Transmission of raw bits over communication channel
- III. Handling user interfaces
- IV. Control and monitoring of subnet
- V. Transmission data through connection oriented or connection less using datagrams

Choose the correct answer from the options given below:

- (A) I-II-III-IV-V
- (B) II-III-IV-V-I
- (C) II-I-IV-V-III
- (D) I-II-IV-V-III

- Q85** Choose the correct option describing the features of Artificial neural network

- I. It is essentially machine learning algorithm.
- II. It is useful when solving the problems for which the data set is very large.
- III. They are able to extract features without input from the programmer.
- IV. These are systems modeled on the human brain and nervous system

Choose the correct answer from the options given below:

- (A) All the statements are correct.
- (B) Only II & III are correct.
- (C) Only I & IV are correct.
- (D) All the statements are not correct.

- Q86** Four persons: P, Q, R and S are in police custody and one of them has committed a crime. They



confess as follows:

- A. Person P: Q did it.
- B. Person Q: S did it.
- C. Person R: I did not do it.
- D. Person S: Q lied.

If exactly one of the statements is false, which of the following is the guilty person.

- (A) P
- (B) Q
- (C) R
- (D) S

Q87 Assertion A:

A Raster scan device is a CRT graphic device and can use a television monitor for display

Reason R:

In Raster scan display the picture is composed of a series of dots. These dots are traced out as a series of horizontal lines. Television works in a similar fashion.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is not the correct explanation of A
- (C) A is true but R is false
- (D) A is false but R is true

Q88 An OS follows round-robin scheduling with time quantum of 4 ms. Assuming that the CPU is free now and there are 20 processes waiting in the ready queue, the maximum amount of time that a process waits before getting into the CPU is _____.

- (A) 80 ms
- (B) 76 ms
- (C) 84 ms
- (D) None of the above

Q89 Statement 1:

Given a graph $G=(V, E)$ in which each vertex $v \in V$ has an associated positive weight $w(v)$, we can use linear programming to find the lower bound on the weight of the minimum-weight vertex cover.

Statement 2 :

The lower bound can be found by maximizing the following

$$\sum_{v \in V} w(v)x(v)$$

subject to

$$x(u) + x(v) \geq 1 \text{ for each } (u, v) \in V$$

$$x(v) \leq 1 \text{ for each } v \in V$$

$$x(v) \geq 0 \text{ for each } v \in V$$

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect and Statement II is correct

Q90 A TCP Server application is programmed to listen on port P on Host S. A TCP Client is connected to the TCP Server over the network. Considered that while TCP Connection is active the server is crashed and rebooted. Assume that the client does not use TCP keepalive timer. Which of the following behaviours is/are possible?

Statement I:

If client is waiting to receive a packet, it may wait indefinitely

Statement II:

If the client sends a packet after the server reboot, it will receive the FIN segment

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Directions (91-95) Read the following passage and answer the given questions.

An organization is granted the block 130.56 .0 .0 / 16. The administrator wants to create 1024 subnets.

Q91 Find the number of addresses in each subnet

- (A) 32
- (B) 64
- (C) 128
- (D) 16

Q92 Find the subnet prefix.

- (A) 130.56
- (B) 136.0
- (C) 136.255
- (D) 136.56 .255

Q93



Find first and last addresses of First subnet.

- (A) 130.56 .0 .0 & 130.56 .254 .254
 (B) 130.56 .0 .0 & 130.56 .0 .63
 (C) 130.0 .0 .0 & 130.255 .255 .255
 (D) 130.56 .0 .0 & 130.56 .255 .63

Q94 Find subnet mask.

- (A) 255 .255 .255.192
 (B) 130.56 .255 .255
 (C) 130.56 .0 .255
 (D) 130.56 .155 .192

Q95 Find first and last addresses of last subnet.

- (A) 130 .56 .0 .0 & 130.56 .254 .255
 (B) 130 .56 .255 .0 & 130.56 .255 .255
 (C) 130 .56 .0 .192 & 130.255 .255 .255
 (D) 130 .56 .255 .192 & 130.56 .255 .255

Directions (96-100) Read the following passage and answer the given questions.

A Turing Machine for the language

$L = a^n b^m c^n d^m | n \geq 1, m \geq 1$ is designed. The resultant model is $M = (\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_f\}, \{a, b, c, d\}, \{a, b, c, d, X_1, X_2, Y_1, Y_2\}, \delta, q_0, B, \{q_f\})$ and part of ' δ ' is given in the transition table. You need to write the following questions based on design of Turing Machine for the given language. Note that, while designing the Turing Machine X_1 and X_2 are used to work with 'a's and 'c's and Y_1 and Y_2 are used to handle 'b's and 'd's of the given string.

	a	b	c	d	X_1	X_2	Y_1	Y_2	B
q_0	(q_1, x_1, R)				M_2				
q_1	(q_1, a, R)	(q_1, b, R)	M_1			(q_1, x_2, R)			
q_2	(q_2, a, L)	(q_2, b, L)			($q_2, X_1,$	(q_2, x_2, L)			

				(R)				
q_3	M_3	(q_4, Y_1, R)			(q_6, x_2, R)			
q_4		(q_4, b, R)	(q_5, Y_2, L)		M_5		(q_4, Y_2, R)	
q_5		(q_5, b, L)			(q_5, x_2, L)	M_4	(q_5, Y_2, L)	
q_6					(q_6, x_2, R)		(q_7, Y_2, R)	
q_7							(q_7, Y_2, R)	(q_f, B, R)

Q96 What is the Move in the cell with number 'M1' of the resultant Table?

- (A) (q_2, X_2, R) (B) (q_2, X_2, L)
 (C) (q_3, X_2, L) (D) Error Entry

Q97 What is the Move in the cell with the number 'M2' of the resultant Table?

- (A) (q_1, X_1, R)
 (B) (q_1, a, R)
 (C) (q_2, X_1, R)
 (D) Error entry

Q98 What is the Move in the cell with number 'M3' of the resultant Table?

- (A) (q_1, x_1, L) (B) (q_4, X_1, R)
 (C) (q_1, x_1, R) (D) Error Entry

Q99 What is the Move in the cell with number 'M4' of the resultant Table?

- (A) (q_5, Y_1, L) (B) (q_3, Y_1, R)
 (C) (q_4, Y_1, L) (D) Error entry

Q100 What is the Move in the cell with number 'M5' of the resultant Table?

- (A) (q_4, X_2, R) (B) (q_5, X_2, R)
 (C) (q_5, X_2, L) (D) Error Entry



Answer Key

Q1	(B)	Q34	(B)
Q2	(A)	Q35	(B)
Q3	(C)	Q36	(C)
Q4	(B)	Q37	(A)
Q5	(B)	Q38	(D)
Q6	(C)	Q39	(A)
Q7	(A)	Q40	(D)
Q8	(B)	Q41	(D)
Q9	(B)	Q42	(B)
Q10	(C)	Q43	(D)
Q11	(A)	Q44	(A)
Q12	(D)	Q45	(D)
Q13	(B)	Q46	(D)
Q14	(B)	Q47	(A)
Q15	(B)	Q48	(D)
Q16	(C)	Q49	(A)
Q17	(B)	Q50	(A)
Q18	(C)	Q51	(D)
Q19	(D)	Q52	(B)
Q20	(B)	Q53	(C)
Q21	(D)	Q54	(A)
Q22	(B)	Q55	(D)
Q23	(A)	Q56	(C)
Q24	(B)	Q57	(B)
Q25	(D)	Q58	(C)
Q26	(A)	Q59	(C)
Q27	(B)	Q60	(B)
Q28	(D)	Q61	(B)
Q29	(C)	Q62	(A)
Q30	(C)	Q63	(C)
Q31	(A)	Q64	(C)
Q32	(B)	Q65	(C)
Q33	(D)	Q66	(D)



Q67 (C)
Q68 (B)
Q69 (C)
Q70 (B)
Q71 (C)
Q72 (B)
Q73 (B)
Q74 (B)
Q75 (D)
Q76 (A)
Q77 (D)
Q78 (D)
Q79 (C)
Q80 (B)
Q81 (C)
Q82 (C)
Q83 (A)

Q84 (C)
Q85 (A)
Q86 (B)
Q87 (A)
Q88 (B)
Q89 (C)
Q90 (C)
Q91 (B)
Q92 (A)
Q93 (B)
Q94 (A)
Q95 (D)
Q96 (B)
Q97 (D)
Q98 (C)
Q99 (D)
Q100 (A)



Hints & Solutions

Q1 Text Solution:

The original statement, "Some students like hockey," is an example of a quantified statement in logic.

The quantifier "Some" refers to the existence of at least one student who likes hockey. In symbolic logic, this statement can be represented as $\exists$ (there exists) student (S) such that they like hockey H.

Now, when we negate this statement, we are essentially saying that it is not true that there exists a student who likes hockey.

In symbolic logic, the negation would be $\neg S(H)$, which is equivalent to $\forall S(\neg H)$, where for all denotes "for all" or "every."

So, the negation "Every student dislikes hockey" is expressing that for every student, it is not the case that they like hockey.

So option B is the correct answer.

Q2 Text Solution:

An equivalence relation on a set x is a subset of $x \times x$, i.e., a collection R of ordered pairs of elements of x , satisfying certain properties. Write " $x R y$ " to mean (x,y) is an element of R , and we say " x is related to y ," then the properties are:

1. Reflexive: $a R a$ for all $a \in R$,
2. Symmetric: $a R b$ implies that $b R a$ for all $a, b \in R$
3. Transitive: $a R b$ and $b R c$ imply $a R c$ for all $a, b, c \in R$.

An partial order relation on a set x is a subset of $x \times x$, i.e., a collection R of ordered pairs of elements of x , satisfying certain properties. Write " $x R y$ " to mean (x,y) is an element of R , and we say " x is related to y ,"

then the properties are:

1. Reflexive: $a R a$ for all $a \in R$,
2. Anti-Symmetric: $a R b$ and $b R a$ implies that for all $a, b \in R$
3. Transitive: $a R b$ and $b R c$ imply $a R c$ for all $a, b, c \in R$.

A total order relation of a set x is a subset of $x \times x$, i.e., a collection R of ordered pairs of elements of x , satisfying certain properties.

Write " $x R y$ " to mean (x,y) is an element of R , and we say " x is related to y ,"

then the properties are:

1. Reflexive: $a R a$ for all $a \in R$,
2. Anti-Symmetric: $a R b$ implies that $b R a$ for all $a, b \in R$
3. Transitive: $a R b$ and $b R c$ imply $a R c$ for all $a, b, c \in R$.
4. Comparability: either $a R b$ or $b R a$ for all $a, b \in R$.

As given in the question, a relation R is defined on ordered pairs of integers as follows: $(x,y) R(u,v)$ if $x < u$ and $y > v$, reflexive property is not satisfied here, because there is $>$ or $<$ relationship between (x,y) pair set and (u,v) pair set.

Other way, if there would have been $x \leq u$ and $y \geq v$ (or $x=u$ and $y=v$) kind of relation among elements of sets then reflexive property could have been satisfied. Since reflexive property is not satisfied here, so given relation can not be equivalence, partial order or total order relation. So, option (A) is correct.

Q3 Text Solution:

Explanation

You and your spouse attend a party with **3 other married couples**.

So total couples = 4

Total people = **8**

Step 1: Maximum Handshakes

Each person:

- Cannot shake hands with themselves
- Cannot shake hands with their spouse

So each person can shake hands with at most **6 people**.

Step 2: Different Answers Condition

You asked 7 people (including your spouse).

Everyone gave a **different answer**.

So possible answers must be:

0, 1, 2, 3, 4, 5, 6

Step 3: Important Logic (Pairing Rule)

Since partners cannot shake hands with each other:

- The person who shook **0 hands** must be married to the one who shook **6 hands**



- The person who shook **1 hand** must be married to the one who shook **5 hands**
- The person who shook **2 hands** must be married to the one who shook **4 hands**

These make 3 couples:

- (0,6)
- (1,5)
- (2,4)

Now only **3** are left.

That unpaired number must belong to your spouse.

Final Answer:

Option C — 3

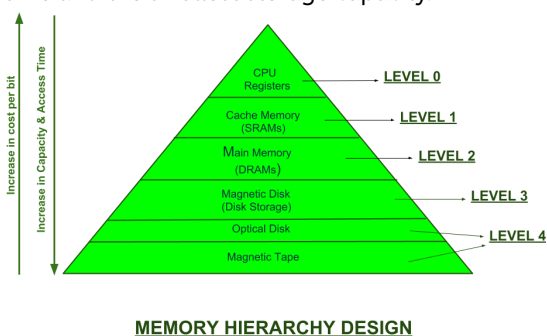
Q4 Text Solution:

- If both x and y are greater than 25, it returns false.
- If both x and y are less than or equal to 25, it returns true.
- If x is greater than 25 and y is less than or equal to 100, it returns false.
Otherwise (i.e., if none of the above conditions match), it returns true

if we take option A , $x > 25$ it will return false; if we take option B $x \leq 25$ it will return true; if we take option C it will return false; if we take $y \leq 100$ then $x \leq 25$ for true value.
So option B is the correct answer.

Q5 Text Solution:

Option A: Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory.
Option B: Registers are small, high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity.



Option C: Main memory, also known as RAM (Random Access Memory), is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower.

Option D: Secondary storage, such as hard disk drives (HDD) and solid-state drives (SSD), is a non-volatile memory unit that has a larger storage capacity than main memory.

So option B is the correct answer.

Q6 Text Solution:

In a basic memory unit (like RAM), the number of address lines determines the number of memory locations, and the number of data lines determines the size of each memory location.

- The number of memory locations is given by 2^n , where n is the number of address lines. This is because each address line can be either 0 or 1 - two possibilities. So with n lines, we have 2^n possible combinations, each representing a different memory location.

- The size of each memory location is determined by m, the number of data lines. Each data line can carry one bit of data. So, m data lines can carry m bits of data.

Combining these two considerations, the total size of the memory is the number of memory locations multiplied by the size of each location in bits.

So, total memory size = 2^n memory locations m bits/location.

So option C is the correct answer.

Q7 Text Solution:

The Karnaugh map (K-map) is a method of simplifying Boolean algebra expressions. The Karnaugh map reduces the need for extensive calculations. Karnaugh map can be explained as An array that contains 2^k cells, where k is the number of variables in the Boolean expression that is to be reduced or optimized.

To simplify the given Boolean function $F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$ using Karnaugh Map (K-Map), let's first construct the K-Map:

- For SOP put 1's in blocks of K-map respective to the minterms (0's elsewhere).
- Make rectangular groups containing total terms in power of two like 2, 4, 8 ..(except 1)



and try to cover as many elements as we can in one group.

- From the groups made in the above step find the product terms and sum them up for SOP form

	C'D' 00	C'D 01	CD 11	CD' 10
A'B' 00	1	0	0	1
A'B 01	0	1	1	0
AB 11	0	1	1	0
AB' 10	1	0	0	1

From Ist group = A'B'C'D' +

AB'C'D'+A'B'CD'+AB'CD'

= B'C'D' + B'CD'

= B'D' -----(1)

From IInd = A'BC'D+A'BCD+ABC'D+ABCD

= A'BD+ABD

= BD -----(2)

So from eq (1) and (2)

F(A, B, C, D)= BD+B'D'

So option A is the correct answer.

Q8 Text Solution:

To calculate the speedup observed in the instruction execution rate when a pipelined machine is considered, let's first analyze the unpipelined machine.

In the unpipelined machine:

- ALU operations take 4 clock cycles (40% of the time).
- Branch operations take 4 clock cycles (20% of the time)
- Memory operations take 5 clock cycles (40% of the time).

Total clock cycle = $0.40 * 4 + 0.20 * 4 + 0.40 * 5$

Total Clock Cycle = $1.6 + 0.8 + 2$

Total Clock Cycle = 4.4 Clock Cycle

Here,

Without using pipeline processor one instruction takes = $4.4 \text{ clock cycle} * 10 \text{ nsec} = 44 \text{ nsec}$

Pipeline time = Clock time + overhead time = $10 \text{ nsec} + 1 \text{ nsec} = 11 \text{ nsec}$ (Clock cycle time is 10 ns

with an additional 1 ns overhead, making it 11 ns

per cycle.)

Speedup = $\frac{\text{Time taken by non-pipeline machine}}{\text{Time taken by pipeline machine}}$

Speedup = $44 / 11 = 4$

So option B is the correct answer.

Q9 Text Solution:

Machine level language, also known as machine code or machine language, is a low-level programming language that directly represents instructions and data in a form that can be understood and executed by a computer's central processing unit (CPU).

In machine level language, instructions are typically represented as binary patterns (0s and 1s) that correspond to specific operations and memory addresses. Each instruction performs a specific task at the hardware level, such as arithmetic or logical operations, data movement, or control flow.

So option B is the correct answer.

Q10 Text Solution:

Option A: The number is always positive when the qualifier 'unsigned' is used. This is correct.

When a variable is declared as 'unsigned', it means that it can only represent non-negative values, including zero. This is because the sign bit is not reserved, and all bits in the variable's memory representation are used to represent the magnitude of the value. So, unsigned variables can only hold positive values or zero.

Option B: The number can be positive or negative when the qualifier 'signed' is used.

This is correct. When a variable is declared as 'signed', it means that it can represent both positive and negative values. The variable reserves one bit for the sign, usually the leftmost bit (the most significant bit). If this bit is 0, the number is positive, and if it's 1, the number is negative. The rest of the bits represent the magnitude of the value.

Option C: The range of values for signed data types is more than that of unsigned data types.

This statement is **incorrect**. The range of values for signed data types is typically less than that of unsigned data types. For example, an 8-bit unsigned data type can represent values from 0 to 255, whereas an 8-bit signed data type can



represent values from -128 to 127. This is because in signed data types, one bit is used to represent the sign, reducing the number of bits available to represent the magnitude.

Option D: The left most bit in the unsigned data type is used to represent the value. This statement is correct because unsigned numbers do not preserve a bit for a sign, so all the bits are used to represent the value of the number. So Option C is the correct answer.

Q11 Text Solution:

A constructor in C++ is a special method that is automatically called when an object of a class is created. To create a constructor, use the same name as the class, followed by parentheses (): Constructors can also be defined outside the class. First, declare the constructor inside the class, and then define it outside of the class by specifying the name of the class, followed by the scope resolution:: operator, followed by the name of the constructor (which is the same as the class). e.g.

```
Date::Date(int dd) { /* ... */ }
```

So Option A is the correct answer.

Q12 Text Solution:

Explanation

Option A: When a user quits a browser

Normally, the session ends if:

No persistent cookie

The browser deletes the session cookie

So the session usually does NOT continue.

Option B: When the user logs out and session is invalidated

When a servlet calls the session.invalidate()

The session is destroyed immediately.

So the session does NOT continue.

Option C: When the session times out due to inactivity

A timeout means the server automatically destroys the session.

So the session does NOT continue.

Option D: When the user refreshes the browser, and there is a persistent cookie

- Refreshing does NOT destroy the session.
- Persistent cookie keeps session ID stored.
- The server recognizes the same session.

So the session continues.

Final Answer:

Option D — When the user refreshes the browser, and there is a persistent cookie

Q13 Text Solution:

Explanation

What is Overfitting?

Overfitting happens when:

- The model learns the training data **too well**
- But performs **poorly on new (test) data**

It memorizes instead of generalizing.

Analyze Each Option

Option A

Training error ↓ and Test error ↓

- Model is learning correctly
- This is **Good learning**, NOT overfitting

Option B

Training error ↓, but Test error ↑

- The model fits the training data very well
- But performance on unseen data becomes worse

This is exactly **Overfitting**

Option C

Training error ↑ and Test error ↑

- The model is not learning properly
- This is **Underfitting**, not overfitting

Option D

Training error and Test error constant

The model is not improving

Not overfitting

Final Answer:

Option B — Training error decreases, but test error increases

Q14 Text Solution:

Option A: A candidate key is a minimal set of one or more attributes that, taken collectively, allows us to uniquely identify any entity in the entity set. This statement is correct. A candidate key is a set of attributes that uniquely identify each tuple in a relation and is minimal in the sense that removing any attribute from the set would cause it to no longer be a candidate key.



Option B: A candidate key for which no proper subset is also a candidate key is called a super key. This statement is incorrect. A super key is a set of one or more attributes that uniquely identify any entity in the entity set. A candidate key is a subset of a super key. Therefore, a candidate key is a super key, but not all super keys are candidate keys.

Option C: A super key is a set of one or more attributes that, taken collectively, allows us to uniquely identify any entity in the entity set. This statement is correct. A super key is a set of attributes that uniquely identifies each tuple in a relation. It may contain additional attributes besides those required to form a candidate key.

Option D: A super key for which no proper subset is also a super key is called a candidate key. This statement is correct. A candidate key is a super key for which no proper subset is also a super key. In other words, a candidate key is a minimal super key, meaning it cannot be further reduced without losing the uniqueness property. So option B is the correct answer.

Q15 Text Solution:

- A primary key is a column or a set of columns in a database table that uniquely identifies each row in the table.
- A foreign key is a column or a set of columns in a database table that establishes a link between data in two tables. It represents a relationship between the tables by referencing the primary key or a unique key in another table.

Given the relations:

EMP (eid, ename, did)

DEPT (did, dname)

The foreign key(s) in the EMP relation would reference the primary key(s) in the DEPT relation. Since the DEPT relation's primary key is the 'did' attribute, any attribute in the EMP relation that references 'did' can be considered a foreign key. Therefore, the possible foreign key(s) in the EMP relation are:

did (which directly references the primary key 'did' in the DEPT relation)

So option B is the correct answer.

Q16 Text Solution:

Explanation:

A functional dependency, denoted $X \rightarrow Y$, refers to a relationship between two sets of attributes. If $X \rightarrow Y$, this means that for all tuples in a relation, if two tuples have the same value for the set of attributes X, they must also have the same value for the set of attributes Y. Similarly, for $Y \rightarrow X$, if two tuples have the same Y value, they must have the same X value.

X	Y	Z
1	1	a
1	1	a
2	1	b
3	2	b

Option A: $X \rightarrow Y$, $Z \rightarrow X$: X is functionally dependent on Y because it uniquely identifies Y value but Z does not hold because for the same value of b, two different values in X are there. So Z is not functionally dependent on X, so **it is incorrect**.

Option B: $Y \rightarrow Z$, $Z \rightarrow X$: Y does not determine Z uniquely because, for 1 in Y, there are two values in Z, so **it is also incorrect**.

Option C: $X \rightarrow Y$, $X \rightarrow Z$: **Y is functionally dependent on X** (because $X \rightarrow Y$). Because it uniquely identifies the Y value, X is functionally dependent on Z because it uniquely identifies the Z value. So, **both functional dependencies are correct**.

Option D: $Y \rightarrow X$, $X \rightarrow Z$: Y does not determine X uniquely because for value 1 in Y, there are two values in X. **So it is also incorrect**.

Therefore, option C is the correct answer.

Q17 Text Solution:

fork():

The fork() system call in Linux is used to create a new process (child process) by duplicating the calling process (parent process). However, fork() does not accept any flags to determine the level of sharing between the parent and child processes. Instead, the child process inherits a copy of the parent process's address space, file descriptors, and other resources.

clone():

The clone() system call in Linux is similar to fork() but provides more flexibility. It allows the caller



to specify various flags to control the level of sharing between the parent and child tasks (processes or threads). With `clone()`, the caller can specify which resources are shared between the parent and child tasks, including the memory address space, file descriptors, and other resources.

`pthread(); pthread()` It allows a program to control multiple different flows of work that overlap in time. Each flow of work is referred to as a thread, and creation and control over these flows is achieved by making calls to the POSIX Threads API.

There is no specific system call named `thread()` in Linux for creating threads.

So option B is the correct answer.

Q18 Text Solution:

In the mid-1980s, researchers at Carnegie Mellon University developed an operating system called Mach that modularized the kernel using the microkernel approach.

- This method structures the operating system by removing all nonessential components from the kernel and implementing them as system and user-level programs.
- Microkernel provides minimal process and memory management, in addition to a communication facility.
- The main function of the microkernel is to provide communication between the client program and the various services that are also running in user space.
- The client program and service never interact directly. Rather, they communicate indirectly by exchanging messages with the microkernel.
- One benefit of the microkernel approach is that it makes extending the operating system easier. All new services are added to user space and consequently do not require modification of the kernel.

So option C is the correct answer.

Q19 Text Solution:

Given an operating system capable of loading and executing a single sequential user process at a time. So, all the requests to disk scheduler will be in sequence and each one will block the execution, hence there is no use of any disk

scheduling algorithm and the OS never has a choice to pick an IO from multiple IOs as there is always one IO at a time. Any disk scheduling algorithm gives the same input sequence and hence the improvement will be 0% for SSTF over FCFS.

So option D is the correct answer.

Q20 Text Solution:

The first come first serve scheduling algorithm states that the process that requests the CPU first is allocated the CPU first.

Since there is no arrival time given for the processes. Waiting time of current Process = Waiting time of previous process + Burst time of the previous Process

Process	Burst Time	Waiting Time
P1	8	0
P2	15	8
P3	10	23
P4	7	33

Average waiting time = (sum of all waiting time) / (Number of processes)

$$= (0+8+23+33)/4$$

$$= 64/4$$

$$= 16$$

So option B is the correct answer.

Q21 Text Solution:

One of the most important steps in determining if a system or software project will succeed is requirements analysis. There are two main categories of requirements: functional requirements and non-functional requirements. Functional Requirements: These are expressed as the expected output, the action carried out, and the input to be supplied to the system. These are the user-specified needs that are visible in the end-user product. e.g. The features that we need to design for this system.

Non-Functional Requirements: As per the project requirements, the system needs to meet some quality criteria known as non-functional requirements. like-

- Portability
- Security
- Maintainability



- Reliability
- Scalability
- Performance
- Reusability
- Flexibility

So option D user interaction is not a Non-functional requirement.

Q22 Text Solution:

The Cocomo Model (Constructive Cost Model) is a software project procedural cost estimate model that is frequently used to accurately anticipate the many project-related factors, including size, effort, cost, time, and quality. So Option B is the correct answer.

Q23 Text Solution:

Software maintenance refers to the process of modifying, updating, and managing software after it has been deployed. It encompasses all activities involved in keeping a software system operational, up-to-date, and aligned with changing requirements and environments throughout its lifecycle. Software maintenance typically includes several key activities:

Option A: Adaptive maintenance- Adaptive maintenance involves modifying the software to adapt it to changes in the environment, such as operating system upgrades, hardware changes, or new regulations. The goal is to ensure that the software remains compatible and effective in its operating environment.

Option B: Corrective maintenance- This involves fixing defects or bugs in the software identified during its use. Corrective maintenance aims to address issues that affect the performance, reliability, or functionality of the software.

Option C: Perfective maintenance-Perfective maintenance focuses on improving the software's performance, efficiency, and user experience. It includes enhancements that add new features, optimize existing functionality, or improve the software's usability based on user feedback.

Option D: Preventive maintenance- Preventive maintenance aims to proactively identify and address potential issues before they impact the software's performance or stability. It involves activities such as code refactoring, performance

tuning, and security audits to prevent future problems.

So Modifying the software to match changes in the ever-changing environment is called option A, Adaptive maintenance.

Q24 Text Solution:

Explanation:

Testing the boundary values of valid and invalid partitions is the foundation of boundary value analysis. Boundaries are an area where testing is likely to produce errors since the behavior at the edge of the equivalence partition is more likely to be erroneous than the behavior within the partition.

It looks for input values that are close to the boundary and are more likely to be incorrect. Each partition has its upper and lower bounds, which are also known as the partition's boundary values.

A boundary value for a valid partition is a valid boundary value. A boundary value for an invalid partition is an invalid boundary value.

For each variable we check-

1. Minimum value.
2. Just above the minimum.
3. Nominal Value.
4. Just below the max value.
5. Max value.

So for the n variable to be checked, **A maximum of 4n+1 test cases are required. So Option B is the correct answer.**

Q25 Text Solution:

Explanation:

The nth Catalan number gives the number of different binary trees with n nodes:

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

The Final Answer is Option D

Q26 Text Solution:

$$\text{Given } f(n) = \begin{cases} 0, & n = 1 \\ f\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right), & n > 1 \end{cases}$$

We need to find $f(25)$ we will recursively solve the problem :

$$f(25) = f\left(\left\lfloor \frac{25}{2} \right\rfloor + 1\right) + 1 \text{-----(1)}$$

$$f(12) = f\left(\left\lfloor \frac{12}{2} \right\rfloor + 1\right) + 1 \text{-----(2)}$$

$$f(6) = f\left(\left\lfloor \frac{6}{2} \right\rfloor + 1\right) + 1 \text{-----(3)}$$

$$f(3) = f\left(\left\lfloor \frac{3}{2} \right\rfloor + 1\right) + 1 \text{-----(4)}$$



$$f(1) = 0$$

Putting the value of $f(1)$ in eq.(4) we get $0+1=1$

Putting the value of $f(3)$ in eq.(3) we get $1+1=2$

Putting the value of $f(6)$ in eq.(2) we get $2+1=3$

Putting the value of $f(12)$ in eq.(1) we get $3+1=4$

So $f(25)$ is 4

Since $\log_2(25) = 4$

So option A is the correct answer.

Q27 Text Solution:

Representation Conversion: First, the polynomials $A(x)$ and $B(x)$ are converted from coefficient form to point-value form. This involves evaluating both polynomials at certain points, typically using roots of unity. This step can be done in $O(n)$ time.

FFT Computation: Once the polynomials are in point-value form, the FFT algorithm is applied to each polynomial separately. The Cooley-Tukey FFT algorithm is commonly used, which has a time complexity of $O(n \log n)$ for polynomials of size n , where n is a power of 2.

Pointwise Multiplication: After computing the FFT of each polynomial, the coefficients are multiplied pointwise. This operation has a time complexity of $O(n)$ since there are n coefficients to multiply.

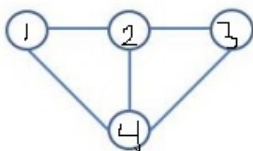
Inverse FFT: Finally, the result of the pointwise multiplication is transformed back to coefficient form using the inverse FFT. Similar to the forward FFT, the inverse FFT also has a time complexity of $O(n \log_2 n)$.

So option B is the correct answer.

Q28 Text Solution:

For a given graph G , a spanning tree can be defined as the subset of G which covers all the vertices of G with the minimum number of edges.

If a graph is a complete graph with n vertices, then total number of spanning trees is $n^{(n-2)}$ where n is the number of nodes in the graph.



Given

The given graph is not complete because each vertex is not connected with the other.

To find spanning trees for the given graph we will follow the given procedure:

STEP 1: Create an Adjacency Matrix for the given graph.

STEP 2: Replace all the diagonal elements with the degree of nodes. For eg. the element at the (1,1) position of the adjacency matrix will be replaced by the degree of node 1, the element at the (2,2) position of the adjacency matrix will be replaced by the degree of node 2, and so on.

STEP 3: Replace all non-diagonal 1's with -1.

STEP 4: Calculate the co-factor for any element.

STEP 5: The cofactor that we get is the total number of spanning trees for that graph.

The Adjacency Matrix for the above graph will be as follows:

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Applying Step 2 and step 3

$$\begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

The matrix can be represented as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

The cofactor of a_{11} of 4×4 matrix is calculated by

$$a_{22}a_{33}a_{44} + a_{23}a_{34}a_{42} + a_{24}a_{32}a_{43} - a_{24}a_{33}a_{42} -$$

$$a_{23}a_{32}a_{44} - a_{22}a_{34}a_{43}$$

$$= 3 \cdot 2 \cdot 3 + (-1)(-1)(-1) + (-1)(-1)(-1) - (-1)(2)(-1) - (-1)(-1)$$

$$(3) - (3)(-1)(-1)$$

$$= 18 - 1 - 1 - 2 - 3 - 3$$

$$18 - 10 = 8$$

Cofactor of (1,1) is 8. Hence total no. of spanning tree that can be formed is option D-8.

Q29 Text Solution:

In a multi-tape Turing machine,

- Each tape has its own head for reading and writing
- Each tape head moves independently of the other heads
- Initially, the input appears on tape 1 and the rest starts out blank



The transition function δ defines how the machine transitions between states based on the current state, the symbols read from each tape, and the symbols to write onto each tape. The transition function δ can be formally defined as follows:

$$\delta : Q \times \Gamma^k \rightarrow Q \times \Gamma^k \times \{L, R, S\}^k$$

$(Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$

- Q, Σ, Γ are all finite sets
 - Q is the set of states
 - Σ is the input alphabet not containing the blank symbol
 - Γ is the tape alphabet
 - $\delta : Q \times \Gamma^k \rightarrow Q \times \Gamma^k \times \{L, R, S\}^k$ is the transition function where k is the number of tapes
 - q_0 is a subset of Q and is the start state
 - q_{accept} is a subset of Q and is the accept state
 - q_{reject} is a subset of Q and is the reject state, where $q_{\text{reject}} \neq q_{\text{accept}}$
- So option C is the correct answer.

Q30 Text Solution:

Let L be a regular language. There exists an integer $p \geq 1$, that depends only on L , such that for every string w in L of length greater than or equal to p , $w = xyz$ (w is the concatenation of strings x , y , and z). For all such w the following conditions are satisfied:

1. $|y| \geq 1$,
2. $|xy| \leq p$, and
3. $xy^n z \in L$ for all $n \geq 0$

In all regular languages any string w that is longer than p , called the pumping length, can be viewed as the concatenation of three strings: x , y , and z , $w = xyz$. Rule 1 tells us that y must exist. Rule 2 gives us a constraint on how long xy can be, and rule 3 states that in addition to xyz being in L , so is $xy^n z$ for all $n \geq 0$, i.e., xz , xyz , $xyyz$, $xyyyx$, etc. The process of repeating the y portion of the string is known as pumping. So the minimum length of 'y' in string 'xyz' is 1. So option C is the correct answer.

Q31 Text Solution:

$A \rightarrow \epsilon$: This notation explicitly states that the non-terminal A can be replaced by the empty string ϵ . It means that when A appears in a

derivation, it can be removed, effectively producing an empty string.

Option A: The production $A \rightarrow \cdot$ indicates that the non-terminal A can be replaced by an empty string, denoted by ϵ .

In the context of LR parsing in the field of computer science, an 'item' of a grammar rule represents a partially recognized rule. The dot in the item is used to indicate how much of a rule has been seen at a particular point in the parsing process.

For example, if we have a rule such as $A \rightarrow B C D$, when parsing begins, no part of the right side has been recognized, so the initial item is $A \rightarrow \cdot B C D$

As parsing proceeds and B , C , and D are recognized, the dot moves rightward: $A \rightarrow B \cdot C D$, $A \rightarrow B C \cdot D$, and $A \rightarrow B C D \cdot$ respectively.

So options B, C, and D are not correct answers.

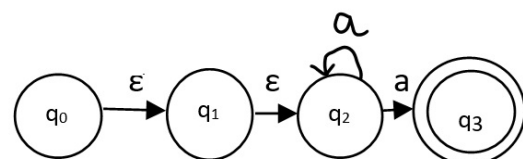
Q32 Text Solution:

Given $M = (\{q_0, q_1, q_2, q_3\}, \{a\}, q_0, \{q_3\})$

Where q_0, q_1, q_2, q_3 are the states, $\{a\}$ is the variable, q_0 is the initial state and q_3 is the final state.

Let's draw the transition diagram for the given table

	ϵ	a
q_0	$\{q_1\}$	$\{\}$
q_1	$\{q_2\}$	$\{\}$
q_2	$\{\}$	$\{q_2, q_3\}$
q_3	$\{\}$	$\{\}$



As we can see to reach the final state q_3 the string is $\epsilon \epsilon a = a$

So the minimum length of string to reach to the final state is 1 which is option B.

Q33 Text Solution:

Given Ethernet address 01011010 00010001 01010101 00011000 10101010 00001111

To convert 8 bit group of data into hexadecimal we will divide 8 bits into two 4 bits data then write corresponding hexadecimal from 0 to 9 and A to F for 10 to 15 respectively.



In hexa decimal the 1st pair of 8-bit data 0101 1010 = 5A
 next, take the second 8-bit pair 0001 0001 = 11
 Similarly 0101 0101 = 55
 similarly 0001 1000 = 18
 similarly 1010 1010 = AA
 similarly 0000 1111 = 0F
 So we will get 5A:11:55:18:AA:0F - Option D.

Q34 Text Solution:

New tokens are added at the rate of r bits/sec which is 2Mbps in the given question.
 Capacity of the token bucket (b) = 16 Mbits
 Maximum possible transmission rate (M) = 10 Mbps
 So, the maximum burst time = $\frac{b}{M-r} = \frac{16}{10-2} = 2$ seconds
 So option B is the correct answer.

Q35 Text Solution:

The delay in Ethernet transmission, which is measured by the duration required to send a single frame over the physical medium, is determined by dividing the frame size by the speed of transmission. In this case, the transmission delay is:
 Transmission delay = Frame size / Transmission rate
 = $512 / 10 \text{ Mbps} = 51.2 \text{ microseconds}$
 In Ethernet, the propagation delay denotes the time duration for a single bit to traverse from source to destination over the physical network medium. It's assessed by dividing the medium's length by the transmission speed of the signal. In this case, the propagation delay is:
 Propagation delay = Length of medium / Propagation speed
 = $2500 \text{ m} / 2 \times 10^8 \text{ m/s} = 12.5 \text{ microseconds}$
 So option B is the correct answer.

Q36 Text Solution:

Cyclic Redundancy Check (CRC) coding involves dividing a data word by a fixed divisor to generate a remainder, which is then appended to the original data word to form the transmitted message.
 In this case, the given data word is 111111, the divisor is 1010, and the remainder is 110.
 To find the transmitted message, you append the

remainder to the original data word. So, the transmitted message is 111111110.
 Therefore, option C is the correct answer.

Q37 Text Solution:

A* problem: Given a network of nodes such that the links between nodes have costs associated with them, find the least expensive path from a starting node to a goal node. (For example, nodes could be cities and link distances.) A* search is an optimal algorithm in that it is guaranteed to find the shortest path if one exists. In addition, A* will be optimally efficient. That is, no other optimal algorithm is guaranteed to expand fewer nodes than A* search. In A* search, node n is expanded with a priority equal to $f(n)$, such that in each cycle of the algorithm, the node with the lowest $f(n)$ value is first expanded, and then removed from the queue. The function $f(n)$ is $f(n) = g(n) + h(n)$ where $g(n)$ is the distance from the start node to node n , and $h(n)$ is a heuristic estimate of the distance from node n to a goal node. If $h(n)$ is always an underestimate of the actual distance from node n to the nearest goal node, then the function h is said to be an admissible heuristic function, and A*'s guarantee of optimal efficiency will hold.
 So option A is the correct answer.

Q38 Text Solution:

The AI technique that enables computers to understand associations and relationships between objects and events is primarily pattern matching.
 Option A) Heuristic Processing: Heuristics are problem-solving techniques that use practical and experience-based methods to quickly arrive at approximate solutions. Heuristics helps in decision-making by finding satisfactory solutions when traditional methods are too slow or impractical.
 Option B) Cognitive Science: Cognitive science is an interdisciplinary field that encompasses the study of how the human mind works, including perception, memory, learning, and problem-solving. It provides insights that AI researchers may use to develop such techniques.



Option C) Relative Symbolism: Symbolic Artificial Intelligence (AI) is a subfield of AI that focuses on the processing and manipulation of symbols or concepts, rather than numerical data.

Option D) Pattern Matching: Pattern matching involves identifying similarities between input data and predefined patterns. AI systems use pattern-matching techniques, such as statistical pattern recognition, machine learning algorithms, and neural networks, to analyze data and their relationships. By recognizing patterns in data, AI systems can make predictions, classify information, and understand relationships between objects and events.

So option D is the correct answer.

Q39 Text Solution:

One optimization method for the minimax algorithm is alpha-beta pruning. It greatly cuts down on computation time. As a result, we may search the game tree considerably more quickly and even reach deeper levels. It removes game tree branches that don't need to be explored because there is already a better move available. Because it feeds two additional parameters—alpha and beta—into the minimax algorithm, it is known as "Alpha-Beta pruning."

The best value the algorithm can promise at that level or higher is called alpha, and the best value it can guarantee at that level or lower is called beta. Alpha and beta are updated continuously as the algorithm searches through the game tree, thereby supplying upper bounds on the values that can be retrieved by the tree's remaining unanalyzed portions. This makes it possible to prune branches that have no chance of influencing the outcome, increasing the algorithm's efficiency.

Alpha and beta values, which are normally initialized to negative and positive infinity, respectively, are introduced in the initial state. Although the end state does offer precise alpha-beta values, these are changed during the search process rather than at the conclusion. So option A is the correct answer.

Q40 Text Solution:

A linear neuron is a type of artificial neuron used in neural networks. The output of a linear neuron

is simply a linear combination of its inputs and weights

To calculate the output of the neuron, we'll use the formula for a linear neuron:

Output = Constant of Proportionality $\times$ (

$$\sum_{i=1}^n weight_i \times input_i$$

Given:

Constant of proportionality = 3

Weights: 1, 2, 3, 4

Inputs: 5, 7, 10, 30

Let's substitute these values into the formula:

$$\text{Output} = 3 \times ((1 \times 5) + (2 \times 7) + (3 \times 10) + (4 \times 30))$$

$$\text{Output} = 3 \times (5 + 14 + 30 + 120)$$

$$\text{Output} = 3 \times 169$$

$$\text{Output} = 507$$

So, the output of the neuron is option D, 507.

Q41 Text Solution:

P: There exists no simple, undirected, and connected graph with 80 vertices and 77 edges. This statement is correct because in a simple, undirected, and connected graph with n vertices, the maximum number of edges is $n-1$. Since 77 edges exceed $80-1=79$

such a graph cannot exist. So, statement P is correct.

Q: All vertices of an Euler graph are of even degree.

This statement is true according to Euler's theorem, which states that an Eulerian circuit exists in a connected graph if and only if every vertex has an even degree. So, statement Q is correct.

R: Every simple, undirected, connected and acyclic graph with 50 vertices has at least two vertices of degree one.

A simple, undirected, connected, and acyclic graph with 50 vertices can be a tree, where all vertices except the leaves have a degree of two. The endpoint of the longest path in the tree can have at least two vertices of degree one.

S: There exists a bipartite graph with more than ten vertices that is 2-colorable.

This statement is true. A bipartite graph is always 2-colorable, meaning its vertices can be divided into two sets such that no two vertices within the same set are adjacent. So, statement S is correct. So among the given statements, P, Q, R, and S are correct, making the total number of correct



statements 4 option D.

Q42 Text Solution:

A. There exists a Boolean algebra with '5' elements.

This is not true. Since a Boolean algebra is made up of all subsets of a set of size n , it has 2^n elements, where n is a positive integer. Since 5 is not a power of 2, it cannot be a Boolean algebra if it has 5 elements.

B. Every element of Boolean algebra has a unique complement.

This statement is true. In a Boolean algebra, each element has a unique complement, which is another element that, when combined with the original element using the Boolean operations (AND, OR, NOT), results in the identity element (0 for OR, 1 for AND).

C. If a Lattice 'L' is a Boolean algebra then 'L' is not relatively complemented.

This is not true. If there is an element b such that $a \wedge b = 0$ and $a \vee b = 1$ for every element a , then the lattice L is said to be complemented. These components are referred to as each other's complements. The lattice is referred to as relatively complemented if these complements are unique. A Boolean algebra is a particular type of lattice in which every element has a unique complement. In other words, 'L' is reasonably complemented if it is a Boolean algebra.

D. The direct product of two Boolean Algebras is also a Boolean algebra.

This statement is true. The direct product of two Boolean algebras is defined as the Cartesian product of their underlying sets equipped with component-wise Boolean operations. The resulting structure satisfies all the properties of a Boolean algebra.

So option B is the correct answer.

Q43 Text Solution:

The main objective for using virtual memory is to increase the effective capacity of the memory system:

This statement is true. Virtual memory allows the computer to use more memory than is physically installed, by using disk space as an extension of RAM. It creates an illusion for programs that they have more memory available than they actually do.

Size of the program can be as large as the size of the secondary memory:

This statement is generally true. Virtual memory systems allow programs to be larger than the physical memory (RAM) available by using disk space as an extension.

Program and data are stored in the secondary memory:

In virtual memory systems, the portions of the program and data that are not active are stored in secondary memory (like a hard drive or SSD).

So option D is the correct answer.

Q44 Text Solution:

1. It is slower than static RAM (SRAM):

This statement is true. DRAM is generally slower than SRAM because it requires periodic refreshing to maintain data, which adds overhead to access times. SRAM, on the other hand, does not require refreshing and typically has faster access times.

2. Packing Density is higher than Static RAM (SRAM):

This statement is true. DRAM typically has a higher packing density than SRAM because it stores each bit as a charge on a capacitor, which requires fewer components compared to the flip-flop design used in SRAM. As a result, DRAM can store more data in the same physical space.

3. It is faster than static RAM (SRAM):

This statement is false. As mentioned earlier, DRAM is generally slower than SRAM due to the need for periodic refreshing. SRAM has faster access times because it uses flip-flops to store data, which retains data as long as power is supplied without the need for refreshing.

4. It requires data refreshing:

This statement is true. DRAM requires periodic refreshing to maintain the charge on its capacitors, which represent the stored data. Without refreshing, the charge on the capacitors would gradually leak away, resulting in data loss. So option A is the correct answer.

Q45 Text Solution:

1. The processor has direct access to both primary and secondary memory:

This statement is false. The processor typically has direct access only to primary memory (RAM),



where actively used instructions and data are stored. Secondary memory (such as hard drives or solid-state drives) is accessed through input/output operations, which are slower than direct memory access.

2. Primary memory stores the active instructions and data for the program being executed on the process:

This statement is true. Primary memory (RAM) is used to store actively used instructions and data for the program currently being executed by the processor. This allows for fast access to the instructions and data needed for computation.

3. Secondary memory is used as a backup memory

This statement is true. Secondary memory can indeed be used for backup purposes. Secondary memory, such as hard disk drives or solid-state drives, primarily serves as long-term storage for data and programs that are not actively used. This storage is persistent, meaning that the data remains even when the computer is powered off. Backup memory refers to a separate storage medium or system used to create copies (backups) of data for protection against data loss due to hardware failures, disasters, or other unforeseen events. These backups can be stored on secondary memory devices.

4. The memory system is implemented on a single-level memory

This statement is false. In modern computer systems, memory hierarchy is typically implemented using multiple levels of memory, such as registers, cache memory, primary memory (RAM), and secondary memory. Each level of memory has different characteristics in terms of speed, capacity, and cost, with the goal of maximizing performance and efficiency. So option D is the correct answer.

Q46 Text Solution:

To understand the effect of scaling on the vector with Cartesian components (x, y) using the given matrix, let's perform matrix multiplication:

$$\begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

The resulting vector will have components:

$$\begin{aligned} \text{New } x &= (0.5 \times x) + (0 \times 1.5) \\ &= 0.5x \end{aligned}$$

$$\begin{aligned} \text{New } y &= (0 \times x) + (1.5 \times y) \\ &= 1.5y \end{aligned}$$

Now, let's analyze the effects based on the resulting components:

Decreases the vertical by three halves: This statement is false. The vertical component (y) is scaled by 1.5, not decreased by three halves.

Increases the vertical by three halves: This statement is true. The vertical component (y) is scaled by 1.5.

Doubles the horizontal: This statement is false. The horizontal component (x) is scaled by 0.5, not doubled.

Halves the horizontal: This statement is true. The horizontal component (x) is scaled by 0.5. So, upon calculation, statements 2 and 4 are correct.

Q47 Text Solution:

Unordered Lists:

Unordered lists are created using the tag and contain a list of items where the order of the items is not important. Each item is typically displayed with a bullet point or some other marker. Example:

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

Ordered Lists:

Ordered lists are created using the tag and contain a list of items where the order of the items is important. Each item is typically displayed with a number or another sequential marker. Example:

```
<ol>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ol>
```

Definition Lists:

Definition lists are created using the <dl> tag and consist of a list of terms and their corresponding definitions. Each term is defined using the <dt> tag, and its definition is provided using the <dd> tag. Example:

```
<dl>
  <dt>Term 1</dt>
```



<dd>Definition 1</dd>

<dt>Term 2</dt>

<dd>Definition 2</dd>

</dl>

Correlated and Enumerated lists

Correlated and Enumerated lists are not a standard HTML concept.

So the correct option is A.

NOTE: The options are modified.

Q48 Text Solution:

Data inconsistency:

File systems do not inherently enforce data consistency. Concurrent access to files by multiple processes or users can lead to data inconsistency issues if proper synchronization mechanisms are not implemented. For example, if one process is writing to a file while another process is reading from it, the reading process might encounter inconsistent or partially updated data.

Difficulty in accessing data:

File systems typically organize data into hierarchical structures (directories and files), which may result in difficulty accessing data efficiently, especially when dealing with large volumes of data. Locating specific files or navigating through directory structures can become cumbersome, particularly in complex file systems.

Data isolation:

File systems lack built-in mechanisms for enforcing data isolation, which can lead to unintended access or modification of data by unauthorized users or processes. Without proper access controls and permissions, sensitive data stored in files may be vulnerable to unauthorized access or tampering.

Atomicity of updates:

File systems do not guarantee the atomicity of updates, meaning that modifications to files or directories may not be performed as a single, indivisible operation. If a failure occurs during a file modification operation (e.g., writing data to a file), it may result in partial updates, leaving the file in an inconsistent or corrupted state.

So option D is the correct answer.

Q49 Text Solution:

1. hierarchical databases: In this, the data is organized into a tree-like structure where each record consists of one parent record and many children. In the hierarchical model, segments pointed to by the logical association are called the child segment and the other segment is called the parent segment. If there is a segment without a parent then that will be called the root and the segment that has no children is called the leaves.

2. In a network database model, data is organized in a more flexible graph-like structure, where records can have multiple connections to other records, forming complex relationships like multiple parent-child relationships. This approach allows for multiple relationships between records, enabling a richer and more versatile representation of data.

3. relational databases-Relational databases represent data using tables consisting of rows and columns, and relationships between entities are established through keys (primary and foreign keys). While relational databases can represent relationships between entities, including one-to-many relationships, they do not inherently enforce a parent-child relationship structure like hierarchical databases.

So option A is the correct answer.

Q50 Text Solution:

Intersection, Union, and Difference operations are basic set operations that are used in many areas of mathematics and computer science, including databases and programming. For the explanation, let's think of sets (or relations, in database terms) as collections of distinct items (or tuples).

- Intersection ($\cap$): The intersection of two sets, R1 and R2, denoted as $R1 \cap R2$, is a set that contains all the tuples that are in both R1 and R2.

here R1

Name	Class
a1	9
b1	5
c1	6
d1	11

R2

Name	Class
------	-------



b1	5
f1	8

$R1 \cap R2$

Name	Class
b1	5

- Union (U): The union of two sets, R1 and R2, denoted as $R1 \cup R2$, is a set that contains all the tuples that are in R1 or R2 or in both.

- Continuing with the previous example, the union of sets R1 and R2

Name	Class
a1	9
b1	5
c1	6
d1	11
f1	8

- Difference (-): The difference operation between two sets, R1 and R2, denoted as $R1 - R2$, results in a set that contains all the tuples that are present in R1 but not in R2.

$R1 - R2$

Name	Class
a1	9
c1	6
d1	11

and $R1 - (R1 - R2)$

Name	Class
b1	5

So options A and C are the correct answers.

Q51 Text Solution:

1. Shortest remaining time first (SRTF) scheduling may cause starvation:

This statement is true. SRTF scheduling prioritizes the process with the shortest remaining execution time, which can lead to longer processes being repeatedly preempted by shorter ones. If there are always shorter processes arriving, longer processes may never get a chance to execute, leading to starvation.

2. Preemptive scheduling may cause starvation:

This statement is true. Preemptive scheduling allows the operating system to interrupt the execution of a process to give access to the CPU

to another process. If there are processes with higher priority constantly arriving, lower priority processes may be repeatedly preempted and may never get a chance to execute, resulting in starvation.

3. Round robin is better than FCFS in terms of response time:

This statement is true. Round-robin scheduling ensures fairness by providing each process with a fixed time slice to execute before being preempted, which helps in achieving better response time compared to first-come, first-served (FCFS) scheduling, especially when there are long-running processes.

So option D is the correct answer.

Q52 Text Solution:

Entry Section (Entry Code):

In this section, a process must request permission or acquire a lock to enter the critical section. Before entering the critical section, the process must ensure that it has exclusive access to the shared resource by using synchronization mechanisms such as locks, semaphores, or mutexes. If the resource is currently being used by another process, the process requesting access to the critical section may need to wait or be placed in a queue until it can acquire the necessary lock.

Critical Section:

This is the section of code where the process performs its critical operations or accesses the shared resource. It is essential that only one process executes this section at any given time to prevent race conditions, data corruption, or inconsistencies.

Exit Section (Exit Code):

After completing its operations in the critical section, the process releases the lock or permission it acquired in the entry section, allowing other processes to enter the critical section if they are waiting. Releasing the lock ensures that other processes can access the shared resource once the current process has finished using it.

Remainder Section (Remainder Code):

This section contains the remainder of the process's code that is not part of the critical section. It includes any non-critical operations or



computations that the process needs to perform outside the critical section.

So option B is the correct answer.

Q53 Text Solution:

1. Requirements gathering must be conducted very carefully:

While requirements gathering is important in any software development process, including agile, being overly meticulous can lead to rigid plans that are difficult to adjust when unpredictability occurs. Agile processes encourage iterative and collaborative requirement gathering, where requirements are refined and adjusted as the project progresses based on feedback and changing priorities.

2. Risk analysis must be conducted before planning takes place:

By understanding risks, teams can develop mitigation strategies and incorporate flexibility so agile methodology performs risk analysis throughout the process not only at the beginning.

3. Software increments must be delivered in short time periods.

Agile methodologies emphasize delivering working software in short, iterative cycles known as iterations. By breaking down the project into smaller increments, teams can quickly adapt to changing requirements, receive feedback from stakeholders, and adjust priorities based on emerging insights, thereby managing unpredictability more effectively.

4. Software processes must not adapt to changes incrementally:

Agile processes advocate for incremental and iterative adaptation to changes throughout the project lifecycle. By embracing change and adjusting processes incrementally, teams can respond more effectively to unpredictable events, evolving requirements, and shifting priorities.

So option C is the correct answer.

Q54 Text Solution:

A. Project Planning: The software project manager is responsible for creating a comprehensive project plan. This includes defining project scope, objectives, timelines,

milestones, and deliverables, and identifying resources needed. Project planning ensures that everyone involved understands what needs to be done, by whom, and by when.

B. Project status tracking: Throughout the project lifecycle, the project manager tracks the progress of various tasks and activities. This involves monitoring milestones, deadlines, resource utilization, and overall project health. Tracking project status helps identify any deviations from the plan early on, enabling the manager to take corrective actions to keep the project on track.

C. Resource management: Managing resources effectively is crucial for the success of a software project. The project manager allocates human, financial, and material resources as per project requirements. This involves tasks such as hiring the right team members, assigning roles and responsibilities, managing workload, and ensuring that resources are utilized efficiently to meet project goals.

D. Risk management: Software projects are inherently risky, and the project manager is responsible for identifying, assessing, and mitigating risks throughout the project lifecycle. This involves identifying potential risks, analyzing their impact and likelihood, developing risk mitigation strategies, and monitoring risk throughout the project. By proactively managing risks, the project manager helps minimize the chances of project failure or delays.

E. Project delivery within time & budget:

Ultimately, the project manager is accountable for delivering the project within the defined time and budget constraints. This involves closely monitoring project progress, managing resources efficiently, controlling costs, and making adjustments as necessary to ensure that the project is completed on time and within budget. So, all the above tasks are managed by the project manager. Therefore, option A is the correct answer.

Q55 Text Solution:

No restriction on redistribution of the software as part or whole: This statement aligns with the principles of open-source software. Open source licenses typically allow anyone to redistribute the



software, whether it's the original version or a modified version. This promotes collaboration and sharing within the software development community.

The integrity of the author's source code must be maintained:

This statement is generally true for open-source software. Most open-source licenses require that if someone distributes modified versions of the software, they must clearly mark those modifications and make the source code of those modifications available. This ensures that the original author's work is respected and that the community can benefit from improvements and modifications.

The software can be sold after distribution: This statement is actually true for most open-source licenses. Open-source software licenses generally do allow the software to be sold.

So, option D is the correct answer.

NOTE: UGC NET has given the correct answer as C, but we find the most appropriate answer as option D.

Q56 Text Solution:

1. The circuit satisfiability problem belongs to class NP: This statement is true. The circuit satisfiability problem can be verified in polynomial time. Given a set of inputs that satisfy the circuit (i.e., make the output 1), we can verify whether the output is 1 by simply traversing the circuit once. If the output is 1, then the inputs satisfy the circuit, and this verification process can be done in polynomial time.
2. The circuit satisfiability problem is at least as hard as any language in NP: This statement is true and reflects the fact that the circuit satisfiability problem is NP-hard. Any problem in NP can be reduced to the circuit satisfiability problem in polynomial time. This means that if we had a polynomial-time algorithm for solving the circuit satisfiability problem, we could use it to solve any problem in NP in polynomial time, making $P = NP$.
3. The circuit satisfiability is NP-Complete: This statement is also true. The circuit satisfiability problem is both in NP (as explained in statement 1) and NP-hard (as explained in statement 2). Therefore, it is NP-complete, meaning it is one of the hardest problems in NP.

4. The size of the circuit is $\theta(K^2+1)$: Suppose A is a decision problem that is solvable in $p(n)$ time by some program P, where n is the length of the input. Also, assume that the input is represented as a sequence of bits. Then, for every fixed n, there is a circuit C_n of size about $O((p(n)^2) \cdot (\log p(n))O(1))$

such that for every input $x = (x_1, \dots, x_n)$ of length n, we have

$$A(x) = C_n(x_1, \dots, x_n)$$

That is, circuit C_n solves problem A on all the inputs of length n.

5. If $P \neq NP$, it wouldn't contradict the existence of NP-complete problems. In fact, NP-completeness arises precisely because we believe that P and NP are different.

When:

$P \neq NP$: This conjecture implies that there exist problems in NP that are not in P, meaning they cannot be solved in polynomial time. These are problems for which there's no known algorithm that can solve them efficiently (in polynomial time) but for which a solution can be verified efficiently (in polynomial time).

NP-Completeness: A problem is NP-complete if it is in NP and every problem in NP can be reduced to it in polynomial time. In other words, an NP-complete problem is at least as hard as the hardest problems in NP.

So option C is the correct answer.

Q57 Text Solution:

1. The prefix notation is $+ / A^*DB - EFG$
This is incorrect. The correct prefix notation for the given expression $(A+B * D) / (E-F)+G$ is: $+ / + A^*B D - E F G$
2. The prefix expression is the reverse of the postfix expression
The postfix expression for the given expression is: $A B D^* + EF-/G +$
This is not correct. If we reverse this, we get $+G /-F E+^* D B A$; This is not the same as the correct prefix notation.
3. The order of operands in infix expression and postfix expression are the same.
This statement is true. In infix notation (the typical mathematical notation), the operands are placed between the operators. In postfix



notation, the order of operands remains the same as in infix notation.

4. The order of operands in infix expression, prefix expression, and postfix expression are the same.

The order of operands in infix expression, prefix expression, and postfix expression are the same" is true. The arrangement of operators varies between these notation formats, and the occurrence of operands remains the same.

So Option B is the correct answer.

Q58 Text Solution:

The given regular expression is $aa^*bb^*cc^*dd^*$, where:

- a^* denotes zero or more occurrences of 'a'
- b^* denotes zero or more occurrences of 'b'
- c^* denotes zero or more occurrences of 'c'
- d^* denotes zero or more occurrences of 'd'

1. The language generated by the regular expression

L

$$= \{a^n b^m c^p d^q \mid n \geq 1, m \geq 1\}$$

$$\{a^n b^m c^p d^q \mid n \geq 1, m \geq 1\}$$

consists of strings where 'a's are followed by 'b's, which are followed by 'c's, and then 'd's.

Each of these letters ('a', 'b', 'c', 'd') can occur one or more times in any order. e.g. abcd, aabccdd... So the language L can generate regular expressions $aa^*bb^*cc^*dd^*$

2. The Context Free Language for the given expression is:

$$S \rightarrow AB|C$$

$$A \rightarrow aAb|ab$$

$$B \rightarrow cBd|cd$$

$$C \rightarrow aCd|aDd$$

$$D \rightarrow bDc|bc$$

This is true. e.g. $S \rightarrow AB|C$

$$S \rightarrow abcd$$

3. The language generated by this expression is equal number of 'a's, followed by equal number of 'b's, followed by equal number of 'c's and followed by equal number of 'd's.

This statement is false. The regular expression $aa^*bb^*cc^*dd^*$ generates strings where each letter ('a', 'b', 'c', 'd') can occur zero or more times,

but there is no constraint that the number of occurrences of each letter must be equal.

For example, the regular expression can generate strings like "abcd", "abcccd", "aaaabbbccddd", etc., where the number of occurrences of each letter is not necessarily equal.

So option C is the correct answer.

Q59 Text Solution:

- Three-address code is an intermediate code. It is used by the optimizing compilers.
- In three-address code, the given expression is broken down into several separate instructions. These instructions can easily translate into assembly language.
- Each Three address code instruction has at most three operands. It is a combination of assignment and a binary operator.
- Basic Block is a set of statements that always executes one after other, in a sequence.

The first task is to partition a sequence of three-address codes into basic blocks. A new basic block is begun with the first instruction and instructions are added until a jump or a label is met. In the absence of a jump, control moves further consecutively from one instruction to another.

The following are the rules used for finding a leader which follows statement 1,3,and 4:

- The first three-address instruction of the intermediate code is a leader.
- Instructions that are targets of unconditional or conditional jump/goto statements are leaders.
- Instructions that immediately follow unconditional or conditional jump/goto statements are considered leaders.

e.g.

Intermediate code to set a 10×10 matrix to an identity matrix:

- 1) $i=1$ //Leader 1 (First statement)
- 2) $j=1$ //Leader 2 (Target of 11th statement)
- 3) $t1 = 10 * i$ //Leader 3 (Target of 9th statement)
- 4) $t2 = t1 + j$
- 5) $t3 = 8 * t2$



```

6) t4 = t3 - 88
7) a[t4] = 0.0
8) j = j + 1
9) if j <= 10 goto (3)
10) i = i + 1           //Leader 4 (Immediately
following Conditional goto statement)
11) if i <= 10 goto (2)
12) i = 1              //Leader 5 (Immediately
following Conditional goto statement)
13) t5 = i - 1         //Leader 6 (Target of 17th
statement)
14) t6 = 88 * t5
15) a[t6] = 1.0
16) i = i + 1
17) if i <= 10 goto (13)

```

There are 6 Basic Blocks in the above code :

- B1) Statement 1
- B2) Statement 2
- B3) Statement 3-9
- B4) Statement 10-11
- B5) Statement 12
- B6) Statement 13-17

Statement -2 :The instruction which is exactly at the middle of the given three address code is a leader. This statement is incorrect. Being in the middle of the code does not inherently make an instruction a leader. Leaders typically mark the beginning of basic blocks, and their determination is based on control flow, not the position within the code.

So option C is the correct answer.

Q60 Text Solution:

- An identity for an operator is a value that when the operator is applied to the identity and some other value, the result is the other value.
- An annihilator for an operator is value that when the operator is applied to the annihilator and some other value, the result is the annihilator.
- For regular expressions we have:
 - I. $\emptyset + L = L + \emptyset = L$
- The expression demonstrates the property that the union of any set L with the empty set is the original set L itself.
- $\emptyset + L$: The symbol "+" typically represents the union operation in language L. So, $\emptyset + L$

represents the union of the empty set ($\emptyset$) and the language "L". Since the union with the empty set results in the same language, this part simplifies to just "L".

- $L + \emptyset$: The result of the expression will be the same as above.
 - II. $\epsilon L = L\epsilon = L$
- This expression pertains to the concatenation operation involving an empty-string ϵ and a language L
- ϵL : An empty string can be thought of as the "zero" of string operations. Concatenating it in the front of any language L does not alter the language, hence the result is L.
- $L\epsilon$: Similarly, concatenating an empty string at the end of any language L also does not change the language, resulting in L.
- $\epsilon L = L\epsilon = L$ means that concatenating either at the start or the end of language L with an empty string ϵ , does not change L.
- III. $\emptyset L = L \quad \emptyset = \emptyset$

This expression pertains to the concatenation operation involving an empty set $\emptyset$ and a language L.

- $\emptyset L$: Concatenating empty set $\emptyset$ with any language L yields an empty set. This is because there are no strings in $\emptyset$ to concatenate with strings in L.
- $L\emptyset$: Similarly, concatenating empty set $\emptyset$ at the end of any language L, also gives empty set, as there are no strings in $\emptyset$ to concatenate with strings in L.
- $\emptyset L = L\emptyset = \emptyset$ means that concatenating either at the start or the end of a language L with an empty set, yields an empty set.
- IV. $\emptyset \cdot L = L \cdot \emptyset = L$
- In the III statement we have proved that that concatenating either at the start or the end of a language L with an empty set, yields an empty set not L. So this statement is false.

Therefore option B is the correct answer.

Q61 Text Solution:

TCP Segment Size = 1400

Packet size = segment size + IP Header Size

= 1400 + 20 = 1420

At R Packet will be Fragmented

MTU = 800 + 20 (Header)



Hence Packet's Data (1400) will be divided into two packets $= (800+600)$

First Fragment $= (800+20) // \text{Data} + \text{Header}$

Second Fragment $= (600+20)$

I: Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.

This statement is correct. Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes and the first fragment 820.

II. If the second fragment is lost then R resends the fragment with an IP Identification value of 0x1234. This statement is false since if any of the fragments are lost then the entire packet has to be retransmitted.

III. If the second fragment is lost then P is required to resend the entire TCP segment.

This statement is true. If any of the Fragments is lost then the entire packet/TCP segment has to be retransmitted.

IV. TCP destination port can be determined by analyzing the second fragment only.

This statement is false. TCP destination port can be determined by analyzing both the fragments not only the second fragment.

So option B is the correct answer.

Q62 Text Solution:

Silly Window Syndrome is a problem that arises due to poor implementation of TCP. It degrades the TCP performance and makes the data transmission extremely inefficient. The problem is called so because:

- It causes the sender window size to shrink to a silly value.
- The window size shrinks to such an extent that the data being transmitted is smaller than TCP Header.

The two major causes of this syndrome are as follows:

- The sender window transmits one byte of data repeatedly.
- The receiver window accepts one byte of data repeatedly.

Cause-1: Sender window transmitting one byte of data repeatedly –

Suppose only one byte of data is generated

by an application. The poor implementation of TCP leads to the transmission of this small segment of data. Every time the application generates a byte of data, the window transmits it. This makes the transmission process slow and inefficient. The problem is solved by Nagle's algorithm.

Nagle's algorithm suggests:

- The sender should send only the first byte on receiving one-byte data from the application.
- The sender should buffer all the rest bytes until the outstanding byte gets acknowledged.
- The sender should wait for 1 RTT(Round Trip Time).

After receiving the acknowledgment, the sender should send the buffered data in one TCP segment. Then, the sender should buffer the data again until the previously sent data gets acknowledged.

- Cause-2: Receiver window accepting one byte of data repeatedly –

Suppose consider the case when the receiver is unable to process all the incoming data. In such a case, the receiver will advertise a small window size. The process continues and the window size becomes smaller and smaller. A stage arrives when it repeatedly advertises a window size of 1 byte. This makes the receiving process slow and inefficient. The solution to this problem is Clark's Solution.

Clark's solution suggests:

- The receiver should not send a window update for 1 byte.
- The receiver should wait until it has a decent amount of space available.
- The receiver should then advertise that window size to the sender.

So option A is the correct answer.

Jacobson's algorithm is used for congestion control by giving dynamic TOT (time-out timer). Piggy backing Algorithm - Piggybacking is the technique of delaying outgoing acknowledgment and attaching it to the next data packet. This is used to improve the efficiency of Network.

Q63 Text Solution:



I. Self-organizing neural network: Self-organizing neural networks, are a type of unsupervised learning neural network. They are used for clustering and dimensionality reduction tasks. It trained its network through a competitive learning algorithm.

II. Backpropagation neural network:

Backpropagation neural networks are a type of supervised learning neural network. They learn by iteratively adjusting the weights of connections between neurons to minimize the difference between predicted and actual outputs. It optimizes the network's performance through gradient descent and error backpropagation.

III. Genetic algorithm: By utilizing biologically inspired operators like mutation, crossover, and selection, genetic algorithms are often used to produce excellent solutions to optimization and search issues. Genetic algorithms mimic the evolutionary process by iteratively generating new candidate solutions and selecting the fittest individuals for reproduction and further evolution.

IV. Forward propagation neural network: Forward propagation neural networks, also known as feedforward neural networks, are a type of neural network where information flows in one direction, from input to output. They are used for various tasks, including classification and regression. Like backpropagation neural networks, forward propagation neural networks do not directly mimic the evolutionary process; instead, they learn from training data using algorithms such as gradient descent.

So option C is the correct answer.

Q64 Text Solution:

The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observations into a number of classes or groups.

I. Logistic regression: Logistic regression is a classification algorithm used to model the probability of a binary outcome based on one or more predictor variables. It is commonly used for

binary classification tasks, where the outcome variable is categorical and has two possible classes.

II. Backpropagation is a learning algorithm used in artificial neural networks for training feedforward neural networks. It is typically used for supervised learning tasks, including both regression and classification.

III. Linear regression: Linear regression is a regression algorithm used to model the relationship between a dependent variable and one or more independent variables. It is used for predicting continuous numeric outcomes rather than categorical classes. Therefore, it is not a classification algorithm.

IV. Forward propagation: Forward propagation is a step in the process of training a feedforward neural network. It involves passing the input data forward through the network's layers to produce an output.

So option C is the correct answer.

Q65 Text Solution:

Linearly separable boolean functions can be solved using a perceptron. If you can draw a line to divide inputs of distinct classes, then the problem is linearly separable.

It can be represented as AND, OR, and NOR as follows:

- AND Operation: Mapping allows linear solutions for each set of inputs (0,0),(0,1),(1,0),(1,1). Two classes—one for outputs that should be 0 and another for outputs that should be 1 which can be created from the points.

- OR Operation: The OR operation is linearly separable, much like the AND operation. It is possible to distinguish between the (1,0),(0,1), and (1,1) inputs that yield 0 and those that yield 1.

- NOR Operation: The NOR operation is linearly separable, just like the OR operation. To put it simply, the NOR operation is the opposite of the OR operation.

But the XOR operation is not linearly separable; that is, the data for 0 and 1 cannot be separated by a straight line. Consequently, a single-layer perceptron cannot represent it.

So option C is the correct answer.

Q66 Text Solution:



A planner graph is a type of graph that is typically used for project planning and scheduling. It frequently shows intricate job dependencies. In graph theory, a graph is said to be "4-Colorable" if it can be colored with a maximum of 4 colors and no two neighboring nodes will have the same color.

b. Bipartite Graphs: These types of graphs are those whose vertices can be split into two sets that are distinct from each other, meaning that no edge joins vertices in the same set. They are frequently employed to depict the connections between two distinct kinds of entities. The graph's vertices can be colored with just two colors according to the "2-Colorable" property, which ensures that no two neighboring vertices have the same color.

c. The Program Evaluation and Review Technique, or PERT, is a project management technique that helps plan, schedule, and oversee the completion of activities. To estimate the time needed to do each work and identify the project's critical path, probabilistic models are used.

d. Critical Path Method (CPM): Another method for managing and scheduling complicated projects is CPM. It is a deterministic model that finds the longest series of dependent jobs along the project network, or the critical route, and utilizes that information to calculate the least amount of time required to finish the project. So option D is the correct answer.

Q67 Text Solution:

In the Assembly programming language, there are different types of address modes:

- Direct Addressing: Also known as absolute addressing, is when the instruction specifies the memory address of the operand.
 - Indirect Addressing: The instruction specifies a register that contains the memory address of the operand.
 - Register Addressing: The instruction specifies the register that contains the operand.
 - Immediate Addressing: The instruction contains the operand.
- Matching each mode with the corresponding example:

- The direct Address would be "STA FFFFH" because the instruction specifies the exact memory address to store an accumulator.
- Indirect Addressing would be "MOV A, M" because the instruction specifies a memory location using an indirect reference which indirectly points to the operand's location.
- Register Addressing would be "MOV A, B" because the instruction defines that the value in register B is to be moved to register A.
- Immediate Addressing would be "MVI A, FFH" because the instruction contains the data directly (in this case, FFH is the immediate data).

So option C is the correct answer.

Q68 Text Solution:

a. If the implementation of generated or derived classes differs only through a parameter we have to use templates.

In C++ templates provide support for generic programming. Template classes and functions eliminate the code duplication of different data types and make development easier and faster.

e.g. We can use more than one generic data type in a class template, and each generic data type is separated by the comma

```
#include <iostream>
using namespace std;
template<class T1, class T2>
class A
{
    T1 a;
    T2 b;
public:
    A(T1 x, T2 y)
    {
        a = x;
        b = y;
    }
    void display()
    {
        std::cout << "Values of a and b are : " <<
a<<" "<<b<<std::endl;
    }
};

int main()
```



```
{
    A<int,float> d(5,6.5);
    d.display();
    return 0;
}
```

Output: Values of a and b are: 5,6.5

b. If the actual types of objects used cannot be known at compile time, then we have to use class hierarchy

Class hierarchy allows us to use the concept of polymorphism. Polymorphism means that a call to a member function will cause a different function to be executed depending on the type of object that invokes the function.

c. If inline operations are essential and templates are used then we have improved run-time efficiency.

Runtime efficiency can be improved by using inline functions and templates because here we need not repeat function calls for different data types used by the same functions or classes.

d. To gain access to differing instances for derived classes through base we have to use explicit casting

Explicit type casting is the process of converting a value from one data type to another data type explicitly. A derived object can be implicitly converted to its base class. In contrast, converting from the base class to the derived class requires an explicit cast operator.

So option B is the correct answer.

Q69 Text Solution:

a. Apache Kafka is an open-source platform used to build real-time streaming data pipelines and applications. It combines messaging, storage, and stream processing to allow storage and analysis of both historical and real-time data. The development of Kafka began at LinkedIn in 2010. The co-founders envisioned a system that could meet LinkedIn's needs through building a real-time infrastructure.

b. Pig Represents Big Data as data flows. Pig is a high-level platform or tool that is used to process large datasets.

c. Apache Mahout is a project of the Apache Software Foundation to produce free implementations of distributed or otherwise

scalable machine learning algorithms focused primarily on linear algebra.

d. Mapreduce is a data processing tool which is used to process the data parallelly in a distributed form.

So option C is the correct answer.

Q70 Text Solution:

a. IPC - Computational speedup:

IPC stands for Inter-Process Communication. It's a mechanism that allows processes to communicate and synchronize their actions. This communication can ensure efficient use of resources, avoid conflicts, and can also lead to computational speedup if communication between processes is done efficiently.

b. Demand Paging - Virtual Memory:

Demand Paging is a strategy for managing computer memory. In this system, only the necessary pages of a program are loaded into physical memory, the rest are kept in virtual memory. This happens on demand, i.e., when a specific page is required.

Virtual Memory is a feature of an operating system (OS) that enables a computer to be able to compensate for shortages of physical memory by transferring pages of data from random access memory (RAM) to disk storage. This process is invisible to the user. Demand Paging is a strategy used in virtual memory systems.

c. Banker's Algorithm - Resource Allocation:

- The Banker's Algorithm is an approach to managing computer resources so that deadlock conditions can be rigorously avoided. It's used in operating systems that manage multiple processes. It helps to determine the safe sequence for allocating resources to the processes.

- Resource Allocation is the assignment of available resources to various uses. In the context of an operating system, these resources include things like CPU time, memory space, and I/O devices. The Banker's Algorithm is one method of managing resource allocation in a way that avoids deadlocks.

d. PCB - Task Control Block:

- PCB stands for Process Control Block. It is a data structure in the operating system kernel containing the information needed to manage a



particular process. It contains details like process state, process identification number, program counter, priority, registers, etc.

- A Task Control Block-as the name implies a similar structure, except it's often used in the context of scheduling and managing "tasks" (essentially processes). So, a Process Control Block (PCB) can be considered a kind of Task Control Block.

So option B is the correct answer.

Q71 Text Solution:

Product complexity:

- Product complexity is related to software complexity. The complexity of any software is measured in terms of software cost estimation.
- There are various cost estimation models used to estimate the complexity of the software. One such model is COCOMO. Product complexity is described as a determinant of software development time.

Structured system analysis:

- It is a software requirement definition. The primary objective of this is to deliver the requirements as specified in the feasibility report.
- It involves accuracy, timeliness, and comprehensiveness of the system output. More clearly defined logical methods for developing system that meet user requirements have led to new techniques that can improve the productivity of analysts and programmers.

Coupling and cohesion: These two are related to software design.

- Software coupling defines how two modules are interdependent on each other. Low coupling is desirable in the system.

- Cohesion describes the intra-dependence of modules. High cohesion is desirable.

Symbolic execution:

- Symbolic execution is a validation technique, or we can say that it is a software testing technique.

- In this, data is represented as symbolic values with expressions. It tests the program by using a flow graph. When testing is done at the end of completion of software then that is known as validation testing.

So option C is the correct answer.

Q72 Text Solution:

- Dijkstra's Algorithm - Stores the total cost from a source node to the current node: Dijkstra's Algorithm is used to find the shortest paths from a source node to all other nodes in a weighted graph. It uses a data structure, often a priority queue, to keep track of the minimum distance from the source node to each node in the graph.

- Prim's Algorithm - Stores minimum cost edge: Prim's Algorithm is a greedy algorithm used to find the minimum spanning tree of a connected, undirected graph. It works by adding the smallest edge that connects a vertex from the growing minimum-spanning tree to a vertex outside the tree.

- Warshall's Algorithm - Calculates path matrix: Warshall's Algorithm is used to find the transitive closure of a directed graph. It determines the reachability between pairs of vertices in a graph and represents the results in the form of a path matrix.

- Kruskal's algorithm - Finds Minimum Spanning Tree: Kruskal's Algorithm is also used to find the minimum spanning tree of a connected, undirected graph. It works by sorting all the edges in ascending order of their weights and adding them to the growing minimum-spanning tree as long as they do not form a cycle.

So option B is the correct answer.

Q73 Text Solution:

List I describes grammar productions while List II describes the lengths of the strings generated by those grammars. Looking at the grammar productions, they generate strings of specific lengths by inserting 'a's at specific positions. Let's interpret each grammar:

- A. ($S \rightarrow aSa \mid aa$), every time you recursively descend, you add an 'a' on both sides of the string. This step doubles the length, starting from the base string 'aa' which has a length of 2.

Therefore, it matches with II. 2^n ; $n \geq 1$. Example: aa, aaaa, aaaaaa,

- B. ($S \rightarrow aaSaa \mid aa$), you add 'aa' on both sides upon each recursive descend, effectively adding 4 to the length of each iteration, starting from the base string 'aa' which has a length of 2. So it generates strings with lengths of $(4n-2)$ for $n \geq 1$ which matches with IV. $2^n - 2$; $n \geq 2$. Example: aa, aaaaaa, aaaaaaa,



- C. ($S \rightarrow \text{aaaSaaa} \mid \text{aa}$), each recursive step adds 'aaa' on both sides, essentially adding 6 to the length on each recursion, starting from 'aa' with length of 2. So it generates strings of lengths $(6n-4)$ for $n \geq 1$, this matches with I. $((2^n * 3)-4)$; $n \geq 1$. Example: aa, aaaaaaaa, aaaaaaaaaaaaaa, ...
- D. ($S \rightarrow \text{aaaaSaaaa} \mid \text{aa}$), each recursion results in 'aaaa' on either side of the string, leading to an increase of 8 characters, starting from 'aa' which has length of 2. So it generates strings of lengths $(8n-6)$ for $n \geq 1$, due to 4 base characters added on each side, this matches with III. $((4 * 2^n)-6)$; $n \geq 1$. Example: aa, aaaaaaaaaa,

So option B is the correct answer.

Q74 Text Solution:

a.CDMA- Code-division multiple access (CDMA) is a channel access method used by various radio communication technologies. CDMA is an example of multiple access, where several transmitters can send information simultaneously over a single communication channel. This allows several users to share a band of frequencies

b.GSM- GSM is a cellular technology that uses combined TDMA and FDMA with a frequency-division duplex for access. Carriers are spaced at 200 KHz and support eight TDMA time slots each. For the up-link, the frequency band 890–915 MHz is allocated, while the down-link uses the band 935–960 MHz.

c.UMTS-UMTS is based on the Global System for Mobile Communications (GSM) standards and was developed by the 3rd Generation Partnership Project. As a third-generation (3G) broadband, packet-based transmission of text, digitized voice, video, and multimedia at data rates up to two megabits per second (Mbps), UMTS offered a vast improvement over its predecessor, 2G mobile technology.

d. WiFi -A wireless network is a computer network that uses wireless data connections between network nodes. Wireless networking allows homes, telecommunications networks, and business installations to avoid the costly process of introducing cables into a building, or as a connection between various equipment locations.

So option B is the correct answer.

Q75 Text Solution:

a.Text Planning: Based on an analysis of the input data, the system decides what details to include in the text that is generated. Making choices regarding the selection of text and arrangement.

b.Sentence Planning: The NLG system arranges the content's natural language expression after it has been decided upon. It chooses the right wording, ordering, and how individual sentences will be structured.

c.Sentence Generation: Using the planned content as a guide, the system generates individual sentences. Choosing the right words, phrases, and syntactic structures is necessary for this. While some NLG systems generate text using pre-defined templates, others might use more advanced techniques like machine learning.

d.Map the input to useful representations: NLU seeks to give machines the ability to comprehend the meaning, context, and intent of human language i.e. ability to convert raw data into the form that a machine can understand. This covers tasks like sentiment analysis, language comprehension, and entity recognition.

So option D is the correct answer.

Q76 Text Solution:

I. There does not exist a polynomial time algorithm to solve integer linear programming problems.

Explanation: This statement is true. Integer Linear Programming (ILP): The ILP problem is given by matrix $A \in \mathbb{R}^{m \times n}$ and vectors $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$. The goal is to find a vector $x \in \mathbb{Z}^n$ such that $A \cdot x \leq b$ and $c^T \cdot x$ is the maximum.

Usually, the problem is given as $\max \{ c^T \cdot x : A \cdot x \leq b, x \in \mathbb{Z}^n \}$.

A large number of practical optimization problems can be modeled and solved using Integer Linear Programming - ILP. The ILP problem differs from the LP problem in allowing integer-valued variables. If some variables can contain real numbers, the problem is called Mixed Integer Programming - MIP. Often MIP is also called ILP, and we will use the term ILP when at least one variable has an integer domain.

If we solve the ILP problem by an LP algorithm and then just round the solution, we could not only get the suboptimal solution, but we can also



obtain a solution that is not feasible. While the LP is solvable in polynomial time, ILP is NP-hard, i.e. there is no known algorithm that can solve it in polynomial time.

II. Main focus of PERT is 'minimizing time'

Explanation : This statement is true. PERT stands for Project Evaluation and Review Technique. This is an appropriate technique that is used for projects where the time required or needed to complete different activities is not known. This planning tool is used to calculate and minimize the amount of time it will take to realistically finish a project.

So option A is the correct answer.

Q77 Text Solution:

Fetch: In the fetch cycle, the CPU retrieves the instruction from memory. The instruction is stored at the address specified by the program counter (PC). The PC is then incremented to point to the next instruction in memory.

Decode: In the decode cycle, the CPU interprets the instruction and determines what operation needs to be performed. This involves identifying the opcode and any operands that are needed to execute the instruction.

Execute: In the execute cycle, the CPU performs the operation specified by the instruction. This may involve reading or writing data from or to memory, performing arithmetic or logic operations on data, or manipulating the control flow of the program.

So option D is the correct answer.

Q78 Text Solution:

do-while loop will execute the code block once, before checking if the condition is true, then it will repeat the loop as long as the condition is true.

Syntax

```
do {
    // code block to be executed
}
```

```
while (condition);
```

The example below uses a do/while loop. The loop will always be executed at least once, even if the condition is false, because the code block is executed before the condition is tested:

Example

```
int i = 0; // initialization is the first step
do {      // execution of the body is
the second step
    printf("%d\n", i);
    i++;
}
while (i < 5); // testing the condition is the third step
```

So option D is the correct answer.

Q79 Text Solution:

Clustering analysis or simply Clustering is basically an Unsupervised learning method that divides the data points into a number of specific batches or groups, such that the data points in the same groups have similar properties and data points in different groups have different properties in some sense. It comprises many different methods based on differential evolution.

In DBSCAN (distance between nearest points) for each point of a cluster, the neighborhood of a given radius has to contain at least a minimum number of points.

- Steps Used In DBSCAN Algorithm
Find all the neighbor points within eps and identify the core points or visited with more than MinPts neighbors.
- For each core point if it is not already assigned to a cluster, create a new cluster.
- Find recursively all its density-connected points and assign them to the same cluster as the core point.

A point a and b are said to be density connected if there exists a point c which has a sufficient number of points in its neighbors and both points a and b are within the eps distance. This is a chaining process. So, if b is a neighbor of c, c is a neighbor of d, and d is a neighbor of e, which in turn is neighbor of a implying that b is a neighbor of a.

- Iterate through the remaining unvisited points in the dataset. Those points that do not belong to any cluster are noise.

So option C is the correct answer.

Q80 Text Solution:



Firmware ROM loads boot block (III): When you turn the power on in a computing system, a built-in piece of software called firmware, which is stored in ROM (Read-Only Memory), is activated first. This firmware includes a program called BIOS (Basic Input/Output System). These programs start the boot process by performing a power-on self-test (POST) to check the health of the system's hardware. Then, they look for a bootable device (like a hard disk, USB drive, etc.). From this device, it loads the boot block, also known as the Master Boot Record (MBR) or GUID Partition Table (GPT) in UEFI systems. This boot block contains information about the layout of the storage device and points to the location of the bootloader.

Bootstrap loader is loaded from the disk (I): The boot block points to the location of the bootloader in secondary storage (hard disk, SSD, USB, etc.). The firmware loads this bootloader, known as a bootstrap loader, into memory. The job of the bootloader is to load and initiate the operating system.

Kernel is loaded onto the memory (II): Once the bootloader is in memory, it loads the operating system kernel into memory. The kernel is the core component of an operating system and interacts directly with the system's hardware. When the kernel is ready, it starts the operating system – bringing us to a state where the system is ready to run applications.

So option B is the correct answer.

Q81 Text Solution:

Assertion A: Cohesion is a qualitative indication of the degree to which a module can be written more compactly and is able to complete its function in a timely manner.

Reason R: Cohesion is a qualitative indication of the degree to which a module is connected to other modules and the outside world.

Explanation: Assertion A correctly defines cohesion as a measure of how well a module can perform its own task. However, Reason R incorrectly defines coupling as a measure of the module's connections to other modules and the outside world.

So Option C - A is correct but R is not correct is the answer.

Q82 Text Solution:

In Kruskal's algorithm, sort all edges of the given graph in increasing order. Then it keeps on adding new edges and nodes in the MST if the newly added edge does not form a cycle. It picks the minimum weighted edge at first and the maximum weighted edge at last.

In option C, edge V is added before edge IV. Here the weight of edge V i.e. 3 is greater than the weight of edge IV i.e. 2 which violates the procedure of the Kruskal algorithm where we need to sort all the edges in increasing order. So option C is the correct answer.

Q83 Text Solution:

Order while simplifying Context Free Grammar is: Elimination of ϵ -productions:- When transforming a Context-free grammar into simplify form, we first remove null-productions, then unit-productions and then useless productions in this exact order. Removing null-productions could give rise to unit-productions, that's why unit is removed after null-productions.

If $A \rightarrow \epsilon$ is a production to be eliminated then we look for all productions, whose right side contains A, and replace each occurrence of A in each of these productions to obtain the non ϵ -productions. These resultant non ϵ -productions must be added to the grammar to keep the language the same.

Example: Remove the null productions from the following grammar

$S \rightarrow ABAC$

$A \rightarrow aA / \epsilon$

$B \rightarrow bB / \epsilon$

$C \rightarrow c$

Solution:

We have two null productions in the grammar $A \rightarrow \epsilon$ and $B \rightarrow \epsilon$. To eliminate $A \rightarrow \epsilon$ we have to change the productions containing A in the right side. Those productions are $S \rightarrow ABAC$ and $A \rightarrow aA$.

So replacing each occurrence of A by ϵ , we get four new productions

$S \rightarrow ABC / BAC / BC$

$A \rightarrow a$

Add these productions to the grammar and



eliminate $A \rightarrow \epsilon$.

$S \rightarrow ABAC / ABC / BAC / BC$

$A \rightarrow aA / a$

$B \rightarrow bB / \epsilon$

$C \rightarrow c$

To eliminate $B \rightarrow \epsilon$ we have to change the productions containing B on the right side. Doing that we generate these new productions:

$S \rightarrow AAC / AC / C$

$B \rightarrow b$

Add these productions to the grammar and remove the production $B \rightarrow \epsilon$ from the grammar.

The new grammar after removal of ϵ – productions is:

$S \rightarrow ABAC / ABC / BAC / BC / AAC / AC / C$

$A \rightarrow aA / a$

$B \rightarrow bB / b$

$C \rightarrow c$

Unit productions:-

A unit production is a production $A \rightarrow B$ where both A and B are non-terminals. Unit productions are redundant and hence should be removed.

Follow the following steps to remove the unit production

Repeat the following steps while there is a unit production

Select a unit production $A \rightarrow B$, such that there exist a production $B \rightarrow \alpha$, where α is a terminal

For every non-unit production, $B \rightarrow \alpha$ repeat the following step

Add production $A \rightarrow \alpha$ to the grammar

Eliminate $A \rightarrow B$ from the grammar

Example: Remove the unit productions from the following grammar

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow C / b$

$C \rightarrow D$

$D \rightarrow E$

$E \rightarrow a$

Solution:

There are 3 unit production in the grammar

$B \rightarrow C$

$C \rightarrow D$

$D \rightarrow E$

For production $D \rightarrow E$ there is $E \rightarrow a$ so we add $D \rightarrow a$ to the grammar and add $D \rightarrow E$ from the grammar. Now we have $C \rightarrow D$ so we add a

production $C \rightarrow a$ to the grammar and delete $C \rightarrow D$ from the grammar. Similarly we have $B \rightarrow C$ by adding $B \rightarrow a$ and removing $B \rightarrow C$ we get the final grammar free of unit production as:

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow a / b$

$C \rightarrow a$

$D \rightarrow a$

$E \rightarrow a$

Useless symbols & productions:- The productions that can never take part in the derivation of any string, are called useless productions. We can see that C, D and E are unreachable symbols so to get a completely reduced grammar we remove them from the CFG. The final CFG is :

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow a / b$

So option A is the correct answer.

Q84 Text Solution:

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

The diagram depicts the seven layers of OSI Model. The Model starts with the Physical layer to the application layer so we will arrange the sequence of working of each layer accordingly which is -

- II. Transmission of raw bits over the communication channel (physical layer)
- I. Transforming the raw bits in the form of a frame for transmission (Data link layer)
- IV. Control and monitoring of subnet (Network layer)
- V. Transmission data through connection-oriented or connection-less using datagrams (Transport layer)
- III. Handling user interfaces (Application layer)



So option C is the correct answer.

Q85 Text Solution:

I. It is essentially machine learning algorithm.

Explanation: This statement is true. An Artificial Neural Network (ANN) is a machine-learning model inspired by the human brain's neural structure. It comprises interconnected nodes (neurons) organized into layers. Data flows through these nodes, adjusting the weights of connections to learn patterns and make predictions.

II. It is useful when solving problems for which the data set is very large.

Explanation: This statement is true. Neural networks can provide robust solutions to problems in a wide range of disciplines, particularly areas involving classification, prediction, filtering, optimization, pattern recognition, and function approximation where the data set is very large.

III. They are able to extract features without input from the programmer.

Explanation: This statement is true. One of its key advantages is the capacity to automatically find the representations required for feature detection or classification from raw data so they are able to automatically extract features without feeding the input by the programmer.

IV. These are systems modeled on the human brain and nervous system

Explanation: This statement is true. The structures and operations of human neurons serve as the basis for artificial neural networks. It is also known as neural networks or neural nets. The input layer of an artificial neural network is the first layer, and it receives input from external sources and releases it to the hidden layer, which is the second layer. In the hidden layer, each neuron receives input from the previous layer neurons, computes the weighted sum, and sends it to the neurons in the next layer. These connections are weighted means effects of the inputs from the previous layer are optimized more or less by assigning different-different weights to each input and it is adjusted during the training process by optimizing these weights for improved model performance.

So option A is the correct answer.

Q86 Text Solution:

Let S 1: P say " Q committed the crime."

S2: Q says "S committed the crime."

S3: R says "I did not do it."

S4: S says "What Q said about me is false."

1. Assume that P is false:

S 1 is false, S 2 is false, S 3 is true, S 4 is true

Since S 3 and S 4 are true, R and S say true. Here more than one statement is false. Therefore, assuming that P is false is not a correct assumption.

2. Assume that Q is false then S1 is true, S3 is true and S4 is true. Here exactly one statement is false. So Q is the guilty person.

3. Assume that R is false:

S 1 is false, S 2 is false, S 3 is false. Here more than one statement is false. Therefore, assuming that R is false is not a correct assumption.

4. Assume that S is false:

S1 is false, S2 is true, S3 is true, S4 is false. Here more than one statement is false. Therefore, assuming that S is false is not a correct assumption.

So option B is the correct answer.

Q87 Text Solution:

- Statement A: "A Raster scan device is a CRT (Cathode Ray Tube) graphic device and can use a television monitor for display."

- This is true since raster scan devices do utilize CRT technology, which is the same technology used in older television monitors. Raster scan devices create images by illuminating individual pixels in an arid (raster), which is similar to the way an old CRT television would display its image.

- Statement R: "In Raster scan display the picture is composed of a series of dots. These dots are traced out as a series of horizontal lines. Television works in a similar fashion."

- This is also true. When producing an image, a raster scan device does so by drawing out a series of horizontal lines composed of individual dots/pixels--this is also known as scanning. This method of drawing an image is similar to older CRT televisions which produce an image through a similar scan process.

So option A is the correct answer.



Q88 Text Solution:

- In round-robin scheduling, every process gets an equal share of CPU time, which is defined by the time quantum or time slice. Assuming there's no context switching time, each process will get to use the CPU for the duration of the time quantum then get placed at the end of the ready queue.
 - In this case, the time quantum is 4 ms and there are 20 processes in total.
 - Since one cycle for all the processes to get CPU once equals to "number of processes" times "time quantum", this means one cycle will take $20 \times 4 = 80$ ms.
 - Given that the CPU is free now, the first process doesn't have to wait, so it gets the CPU immediately. But the last process, i.e., the 20th process has to wait for 19 processes (not the full 20, because the first process didn't wait) to complete their quanta before getting CPU. That means
 - Maximum wait time = (Number of processes - 1) * Time quantum
 - Maximum waiting time for a process is $19 \times 4 = 76$ ms.
- So option B is the correct answer.

Q89 Text Solution:

- Statement 1:
 - Given a graph $G=(V, E)$ in which each vertex $v \in V$ has an associated positive weight $w(v)$, we can use linear programming to find the lower bound on the weight of the minimum-weight vertex cover.
 - Explanation:
 - This statement is correct. Linear programming can be used to find the lower bound on the weight of the minimum-weight vertex cover by formulating it as an optimization problem.
 - Statement 2:
 - The lower bound can be found by maximizing the following
- $$\sum_{v \in V} w(v)x(v)$$
- subject to
- $$x(u) + x(v) \geq 1 \text{ for each } (u, v) \in V$$
- $$x(v) \leq 1 \text{ for each } v \in V$$
- $$x(v) \geq 0 \text{ for each } v \in V$$
- Explanation: This is incorrect. For the minimum

weight vertex cover problem, the objective function should be minimized, not maximized.

So the lower bound can be found by

$$\text{Min } \sum_{v \in V} w(v)x(v)$$

Subject to

$$x(u) + x(v) \geq 1 \text{ for each } (u, v) \in V$$

$$x(v) \in \{0, 1\} \text{ for each } v \in V$$

$$x(v) \geq 0 \text{ for each } v \in V$$

So option C is the correct answer.

Q90 Text Solution:

I: If the client was waiting to receive a packet, it may wait indefinitely.

This option is also correct. There is no Keepalive timer exists, then there is a possibility that the Client may keep waiting Indefinitely for the packet from the server.

II: If the client sends a packet after the server reboot, it will receive the FIN segment.

This option is incorrect.

Because in the Absence of a Keepalive timer, after reboot TCP server does not remember info regarding previous connections, and If it receives a packet from the client, it may generate a reset segment, not a FIN segment.

So option C is the correct answer.

Q91. Text Solution:

$2^n = 1024$ Therefore, $n = 10$. The address given here is a Class B address, therefore the default mask is /16. 10 bits are required for subnets and hence the required subnet mask is $/(16+10) = /26$. In dotted decimal format, it shall be 192, and therefore, the subnet mask will be 255.255.255.192. The remaining bits must be used for the address in each subnet i.e. there will be $32 - 26 = 6$ bits available for the address component. Therefore, there will be a total of $2^6 = 64$ bits available.

So option B is the correct answer.

Q92. Text Solution:

Subnet masks (IPv4) and prefixes (IPv6) identify the range of IP addresses that make up a subnet, or group of IP addresses on the same network. Given 130.56.0.0 / 16 belongs to class B

So option A is the correct answer.

Q93. Text Solution:

- The IP address block is 130.56.0.0 / 16 and the administrator wants to create 1024 subnets,



subnets would have a / 26 subnet mask, meaning there are 64 addresses in each subnet.

- Therefore, the first and last addresses of the first subnet would be:

- The first address (network address) would be the first address of the granted block, that is:

130.56 .0 .0

- Since there are 64 addresses in each subnet, the last address (broadcast address) would be the 63rd increment from the first address, that is: 130.56.0.63.

- So the correct answer would be:

First address =130.56 .0 .0, Last address =130.56 .0 .63 option B.

Q94. Text Solution:

The address given here is a Class B address, therefore the default mask is /16. 10 bits are required for subnets and hence the required subnet mask is $/(16+10) = /26$. In dotted decimal format, it shall be 192 and therefore, the subnet mask will be 255.255.255.192

So option A is the correct answer.

NOTE: Options are wrong in the paper. we have modified the option.

Q95. Text Solution:

The IP address block is 130 . 56 . 0 . 0 / 16 and the administrator wants to create 1024 subnets, subnets would have a / 26 subnet mask, meaning there are 64 addresses in each subnet.

- Therefore, the first and last addresses of the first subnet would be:

- The first address (network address) would be the first address of the granted block, that is:

130.56 .0 .0

- Since there are 64 addresses in each subnet, the last address (broadcast address) would be the 63rd increment from the first address, that is: 130.56.0.63.

First address =130.56 .0 .0, Last address =130.56 .0 .63

Similarly, first and last addresses of last subnet would be 130 .56 .255 .192 & 130.56 .255 .255

So option D is the correct answer.

Q96. Text Solution:

Given A Turing Machine for the language

$L = a^n b^m c^n d^m | n \geq 1, m \geq 1$ is designed

Turing Machine X1 and X2 are used to work with 'a's and 'c's and Y1 and Y2 are used to handle 'b's and 'd's of the given string.

Let us draw the state table for turning machine for the language L with the help of an example of string B a a b b b c c d d d B

- we will start with q0 and get 'a' which is replaced by X1 and move to right direction.
- At q1 if we get 'a' leave it as it is, if we get b leave it as it is and move in the right direction. When we get c, it will be replaced by X2 and go to the left direction to find out last check point.
- At q2 state we will get some 'a' or 'b', leave it as such. whenever we get last checkpoint X1 while moving in left direction, we will start move in right direction.
- At q3 state whenever we get 'a' replace with X1, and merge with state q1 because now the same process will be going on on the way we will get b leave it as it is on the way we will get X2 also whenever we are getting X2 leave it X2 as it is and move to right now whenever we are getting c this c will be replaced with X2 with this we can handle it after that we are going to state q2 and we start moving to the left direction again now whenever we are starting moving to left Direction now we will get this X2 that's why X2 will be X2 and move to left now finally whenever we are able to reach b will be as it is. Now we are able to reach this X1 position it was our last checkpoint.
- At q3 we will get b and will be replaced by Y1 and move to the right direction.
- Now we are at q4 we will get d which is replaced by Y2 and move to the left direction.
- At q5 we will get b and X2 leave it as it is. whenever we get Y1, we will move in the right direction.
- At q6 we will get b which is replaced by Y1 and by moving in the right direction we will be at q4 where we will find Y2.
- Now moving towards the left direction we are at q5 and get Y2.
- At q6 if we are getting X2 leave it as such. if we are getting Y2 to move in the right



direction at the q7 state. here we will get blank state i.e. our final state.

	a	b	c	d	X ₁	X ₂	Y ₁	Y ₂	B
q0	(q1, x1, R)				M2				
q1	(q1, a, R)	(q1, b, R)	(q2, X2, L)			(q1, x2, R)			Q97.
q2	(q2, a, L)	(q2, b, L)			(q3, X1, R)	(q2, x2, L)			Text Solution:
q3	(q1, x1, R)	(q4, Y1, R)							M2 is an error entry. there is no transition.
q4		(q4, b, R)		(q5, Y2, L)		(q4, X2, R)		(q4, Y2, R)	So option D is the correct answer.
q5		(q5, b, L)				(q5, x2, L)	M4	(q5, Y2, L)	Q98. Text Solution:
q6						(q6, x2, R)		(q7, Y2, R)	From the state table we can find out the value of M3 is option C , (q1, x1, R).

q7						(q7, Y2, R)			(qf, B, R)
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So from the state table, we can see that while getting c which is replaced by X2 we will move to q2 in the left direction, so option B (q2, X2, L) is the correct answer.

Q97. Text Solution:

M2 is an error entry. there is no transition.

So option D is the correct answer.

Q98. Text Solution:

From the state table we can find out the value of M3 is option C , (q1, x1, R).

Q99. Text Solution:

So at M4 no entry is there. So option D is the correct answer.

Note: Option is modified.

Q100. Text Solution:

From the state table we can find out the value of M5 is option A , (q4, X2, R)



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